

Maximum Likelihood Neural Additive Models

Abstract

Neural additive models (NAMs) have attracted increasing attention recently due to their promising interpretability and approximation ability. However, existing works on NAMs are typically limited to the mean squared error (MSE) criterion, which can suffer from degraded performance when confronted with data containing non-Gaussian noise, such as outliers and heavy-tailed noise. To address this issue, we utilize maximum likelihood estimation for error modeling and formulate noise distribution-aware additive models, called the *Maximum Likelihood Neural Additive Models* (ML-NAM). It employs kernel density estimation to avoid explicit assumptions about the noise distribution, [allowing it to adapt flexibly to diverse noise environments](#). Theoretically, the excess risk bounds are established for ML-NAM under mild conditions, and the resulting minimax convergence rate exhibits polynomial decay when the target function lies in a Besov space. Empirically, extensive experiments validate the effectiveness and robustness of the proposed ML-NAM [in comparison to](#) several state-of-the-art approaches. Across multiple datasets, ML-NAM practically reduces MSE by 14%-29% compared to NAM under non-Gaussian noise. This work enables reliable decision-making in high-stakes domains where robustness and interpretability are essential to real-world applications.

Keywords: Neural additive model, Robust regression, Error bound, Interpretable machine learning, Kernel density estimation.

1. Introduction

Neural Additive Models (NAMs) (Agarwal et al., 2021) rely on interpretable deep learning architectures, where the data-driven estimator is considered as the sum of component functions, and each component function is approximated by a univariate neural network associated with a single input feature. This structure allows NAMs to overcome “the curse of dimensionality” (Altman and Krzywinski, 2018) and be effectively applied across various

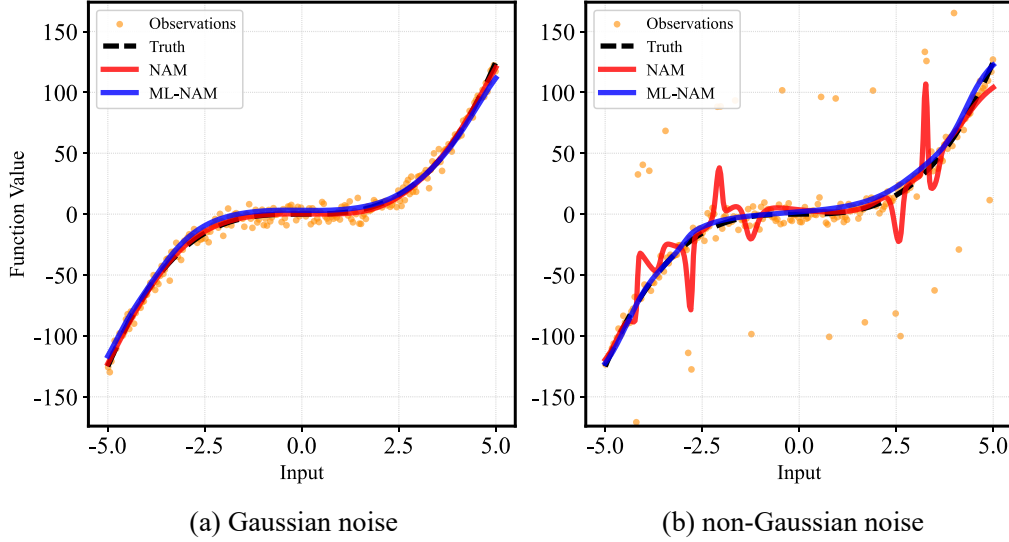


Figure 1: The sensitivity of MSE for NAM with simulation data from $f(x) = x^3$ associated with different noise distributions. (a) Both models perform well under Gaussian noise. (b) With non-Gaussian noise (e.g., Student-t distribution and outliers), MSE-based NAM is sensitive to extreme values, leading to noticeable deviations. In contrast, the proposed model remains robust to heavy-tailed noise and outliers.

8 learning tasks (Zhuang et al., 2021) and high-stakes domains (Berrisch et al.,
9 2023; Condor and Pardos, 2024), i.e., biological analysis and medical diag-
10 nosis (Karim et al., 2023).

11 Due to the strong interpretability and approximation ability of NAMs,
12 various extensions of NAMs have been investigated recently in the machine
13 learning community. Along this line, there are mainly two branches concern-
14 ing the computational feasibility and interpretability under milder [structural](#)
15 [constraints and theoretical assumptions](#), respectively. From the perspective
16 of computing algorithms, NAMs contain multiple neural networks, which
17 may face a heavier computational burden than traditional additive models.
18 To improve computational feasibility, researchers have introduced various
19 constraint mechanisms, [such as parameter sharing](#) (Radenovic et al., 2022;
20 Jo and Kim, 2023) and [high-level concept representations](#) (Duong et al.,
21 2024), which enable NAMs to reduce the number of parameters while main-
22 taining learning performance significantly. In terms of interpretability, the
23 decomposition of the univariate function (Agarwal et al., 2021) ignores the

interactions among input variables. To further enhance the interpretability of NAMs, extensive studies have been conducted in Chang et al. (2022); Yang et al. (2021); Bouchiat et al. (2024); Feddersen and Cleophas (2025), where additive learning architectures incorporate first-order or higher-order interactions of input variables for data-driven structure discovery.

Despite recent advancements in the architecture of NAMs, most existing methods still rely on the least squares loss, which is often derived from the mean squared error (MSE) criterion. This makes NAMs sensitive to non-Gaussian noise (e.g., outliers and heavy-tailed noise), resulting in degraded performance, as illustrated in Figure 1. Indeed, observations are often corrupted by non-Gaussian noise in real-world applications. This motivates us to formulate a new estimation method by replacing the least squares loss with other robust error metrics, e.g., the likelihood-based losses (Li et al., 2021; Wang et al., 2025). Moreover, theoretical clarity for NAMs is largely absent, including guarantees for approximation and generalization. In particular, it remains unclear how their additive structure interacts with neural network components in the presence of complex, non-Gaussian noise. This gap in theoretical insight hinders a comprehensive understanding of NAMs’ robustness and generalization properties.

To advance the understanding and reliability of NAMs in real-world settings, further research is needed to address the following foundational questions:

- *How to reformulate NAM to improve its robustness against non-Gaussian noise?*
- *Can we establish theoretical guarantees for the generalization ability of NAM?*

One promising direction toward addressing these challenges is to adopt Maximum Likelihood Estimation (MLE) for data modeling, which provides a flexible and theoretically robust alternative within statistical machine learning (Immer et al., 2021; Blasques et al., 2022). MLE aims to identify the parameter values that maximize the likelihood of observations under a specified statistical model, and it offers well-established theoretical guarantees such as sufficiency, consistency, efficiency, and parameterization invariance (Gould et al., 2006). In contrast to conventional NAMs that optimize MSE and thus implicitly assume Gaussian noise, we employ MLE directly in the training process to accommodate arbitrary error distributions. Specifically,

the NAM module predicts additive shape functions, while the kernel density estimation (KDE) modeling residuals adaptively (Gizewski et al., 2022). The overall optimization objective maximizes the likelihood of these residuals under the estimated density.

Building on these insights, we formulate new additive models, called *Maximum Likelihood Neural Additive Models* (ML-NAM), for the noise distribution-aware setting. In our framework, the likelihood-based objective aligns model training with the empirical data distribution and adaptively weights observations according to their estimated density. This form can also recover standard regression losses, including the least squares and least absolute deviation losses, which are limited to the assumptions of Gaussian noise and Laplace noise, respectively. The NAM structure further enables flexible modeling of nonlinear relationships while preserving interpretability. To the best of our knowledge, the integration of MLE with NAMs under general noise has not been systematically investigated in the prior literature. As a theoretical complement, we also analyze the generalization properties of ML-NAM under mild smoothness and regularity conditions. In what follows, we present the contributions of this work.

1.1. Summary of Contributions

We develop a robust neural additive model and establish its statistical guarantees under general non-Gaussian noise conditions. Specifically, the main contributions of this work are highlighted below.

- *Robust Model Architecture:* We propose a new class of additive models (called ML-NAM) under a flexible and robust nonparametric framework based on MLE. In contrast to existing approaches that rely on restrictive distributional assumptions, ML-NAM accommodates a wide range of noise structures, including heavy-tailed and heterogeneous distributions, while preserving interpretability. The likelihood-based formulation enables the model to capture the full data distribution (see (Wang et al., 2025)), thereby enhancing robustness and generalization across diverse regression tasks.
- *Theoretical Guarantees:* We derive non-asymptotic upper bounds on the excess risk of the ML-NAM estimator. When the target function belongs to the mixed smooth Besov space, our estimator achieves a minimax convergence rate with polynomial decay under mild conditions

(see Corollary 1). Specifically, our analysis applies to general data settings and does not rely on the assumption of Gaussian noise.

- *Empirical Validation:* Experiments on both simulated datasets and real-world datasets empirically demonstrate ML-NAM’s superior robustness against outliers and non-Gaussian noise, alongside improved accuracy, showcasing its practical efficacy.

1.2. Related Works

As a special class of generalized additive models, NAMs (Agarwal et al., 2021) approximate the target function through an additive set of univariate neural subnetworks, enabling interpretable learning in medical imaging (Kathirvel et al., 2025) and genomics (Li et al., 2025). Despite these advances, two fundamental limitations persist in existing NAM frameworks. Firstly, their universal reliance on MSE implicitly assumes Gaussian noise, rendering models vulnerable to performance degradation in real-world conditions with heavy-tailed distributions or outliers. Secondly, theoretical guarantees remain incomplete, where prior work (Xu et al., 2023) establishes convergence rates only under restrictive assumptions (e.g., linearized networks), failing to characterize the generalization of complete deep neural network models under general noise conditions.

Recent efforts address peripheral challenges: computational efficiency via parameter sharing (Jo and Kim, 2023) or concept embeddings (Duong et al., 2024), interpretability through interaction terms (Yang et al., 2021) or Bayesian inference (Bouchiat et al., 2024), and convex optimization (Zhang et al., 2024). However, these approaches retain the MSE paradigm and do not resolve the core noise-sensitivity limitation. Similarly, while Thielmann et al. (2024) integrates distributional regression, they maintain restrictive assumptions on noise. This reveals a critical research gap: few existing works systematically address robustness to arbitrary noise distributions while providing rigorous theoretical guarantees for complete NAMs.

Our proposed Maximum Likelihood Neural Additive Models (ML-NAM) directly target these limitations. As summarized in Table 1, ML-NAM replaces MSE with maximum likelihood estimation using kernel density estimation to approximate arbitrary noise structures without distributional assumptions. To our knowledge, this work constitutes the first systematic integration of maximum likelihood estimation with NAMs, establishing both

Table 1: Related works on neural additive models ($\checkmark$ represents “Yes” and $\times$ means “No”).

References	Loss function	Global/Sub-network	Generalization guarantee	Robustness
Agarwal et al. (2021)	Least squares	Fully connected neural network	$\times$	$\times$
Chang et al. (2022)	Least squares	Neural oblivious decision trees	$\times$	$\times$
Xu et al. (2023)	Least squares	Fully connected neural network	$\checkmark$	$\times$
Bouchiat et al. (2024)	Log-marginal likelihood	Bayesian neural network	$\times$	$\times$
Duong et al. (2024)	Least squares	Taylor neural network	$\times$	$\times$
Zhang et al. (2024)	Least squares	Random Fourier feature	$\times$	$\times$
Ours	Maximum likelihood	Fully connected neural network	$\checkmark$	$\checkmark$

algorithmic robustness and theoretical foundations for interpretable modeling under general noise conditions.

2. Preliminaries

This section recalls the basic setup of nonparametric regression and specifies the form of the target log-likelihood function. We also detail the architecture of neural additive models and summarize the main notations and their corresponding definitions in Table 2.

2.1. Problem Formulation

Consider a nonparametric regression model

$$y = f^*(\mathbf{x}) + \epsilon, \quad (1)$$

where the input $\mathbf{x} = (x^1, \dots, x^p)^\top \in \mathbb{R}^p$, the response $y \in \mathbb{R}$ and the random noise ϵ is independent of $\mathbf{x}$. Usually, the purpose of regression learning is to approximate the ground truth f^* in terms of the limited sample set

$$\mathcal{Z}_n = \{\mathbf{z}_i := (\mathbf{x}_i, y_i)\}_{i=1}^n \subset \mathbb{R}^p \times \mathbb{R},$$

where each sample is drawn independently from the distribution determined by (1). Under the Gaussian noise assumption, the following least-squares estimator

$$\hat{f}_{\text{MSE}} = \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n (y_i - f(\mathbf{x}_i))^2 \quad (2)$$

is used extensively for regression learning associated with the hypothesis function space $\mathcal{F}$, owing to its simplicity and optimality in this Gaussian noise setting.

Table 2: Overview of the main notations used in this paper.

Notations	Descriptions
f^*	the ground-truth target function
$f_{\mathcal{H}}^*$	the optimal predictor in the hypothesis space $\mathcal{H}$
f_j	the j -th component function of the ML-NAM
$\mathcal{H}(\mathcal{S}, \mathcal{U}, \mathcal{B})$	the ML-NAM hypothesis function space
$\mathcal{B}$	the upper bound of f^* and $f \in \mathcal{H}$
$\mathcal{L}, \mathcal{L}_n$	the population risk and its empirical version associated with ML-NAM
$\hat{f}, \tilde{f}$	the ML-NAM estimators under density-known and density-unknown settings
$h, \hat{h}$	the density function of the residual and its empirical estimator
$d_l^j, \mathbf{v}_l^j, \phi_{\mathbf{v}_l^j}, A_j^{(l)}$	the width (number of units) of layer l , the threshold vector of layer l , the activation function associated with $\mathbf{v}_l^j$ and the weight matrix of layer l in the j -th subnetwork, respectively
L_j, W_j, S_j, B_j	the number of hidden layers, the maximum hidden-layer width, the number of nonzero parameters, and the parameter-norm upper bound of the j -th subnetwork, respectively

148 However, the least squares loss in (2) can be suboptimal in scenarios
149 where data contain outliers or the distribution of ϵ is heavy-tailed, poten-
150 tially leading to unstable estimates. In fact, when estimating the conditional
151 mean function $f(\cdot)$, observations where y_i is closer to $f(\mathbf{x}_i)$ are supposed
152 to contribute more information than those with significant deviations from
153 the conditional mean. This idea can be naturally formulated in terms of the
154 probability of observation. To address this limitation, Wang et al. (2025) pro-
155 poses a loss function built upon an Estimated Maximum Likelihood (EML)
156 function, which takes the following form:

$$\hat{f}_{\text{EML}} = \arg \max_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \log h(y_i - f(\mathbf{x}_i)), \quad (3)$$

157 where h is the density function of the residual $\epsilon_i = y_i - f(\mathbf{x}_i)$. From the stand-
158 point of maximum likelihood estimation, data points with residuals closer to
159 the conditional mean correspond to a higher probability density. Leveraging
160 this property, EML learns the residual density directly from data, providing
161 a flexible nonparametric noise model that avoids explicitly specifying the er-
162 ror distribution and naturally adapts to the underlying data characteristics
163 (Wang et al., 2025; Chen et al., 2024).

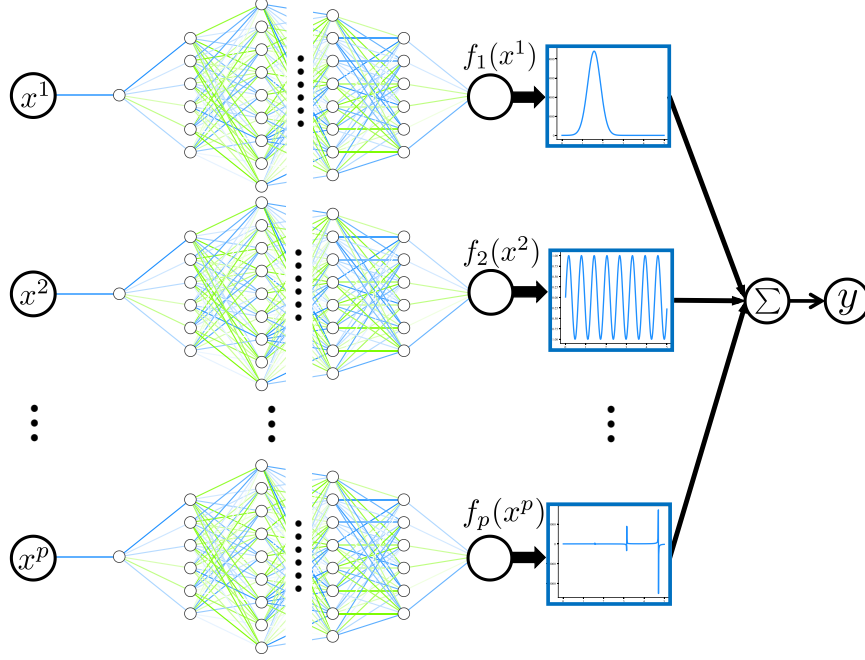


Figure 2: NAMs architecture for regression: each input variable is modeled by a separate neural network that learns its associated shape function. This design enables a direct interpretation of how each feature influences the target variable.

164 2.2. Maximum Likelihood Neural Additive Models

165 Additive models, as natural extensions of linear models, represent the re-
 166 lation between covariates and response variables as the sum of low-dimensional
 167 component functions (Stone, 1985). Usually, additive models provide inter-
 168 pretable predictions and the ability to circumvent the curse of dimensionality
 169 (Hastie, 2017). The hypothesis function in additive models is defined as

$$f(\mathbf{x}) = \sum_{j=1}^p f_j(x^j), \quad (4)$$

170 where $f_j : \mathbb{R} \rightarrow \mathbb{R}$ is the j -th component function and x^j is the j -th feature
 171 of the input $\mathbf{x}$ for $j = 1, \dots, p$.

172 Recently, NAMs (Potts, 1999; Agarwal et al., 2021) have extended clas-
 173 sical additive models by using a small dedicated neural network to represent
 174 each component function f_j in (4) (see Figure 2). Each function f_j provides a
 175 transparent mapping from a single feature to its additive contribution to the

176 prediction, allowing for the direct interpretation of feature-specific effects,
 177 such as linear trends or nonlinear patterns. In this way, the architecture
 178 combines the expressivity of deep neural networks with the inherent intelli-
 179 gibility of generalized additive models. Similar to NAMs in Agarwal et al.
 180 (2021), we define f_j as the j -th subnetwork

$$f_j(x^j) = A_j^{(L_j)} \phi_{\mathbf{v}_{L_j}^j} \circ A_j^{(L_j-1)} \phi_{\mathbf{v}_{L_j-1}^j} \circ \dots \circ A_j^{(1)} \phi_{\mathbf{v}_1^j} \circ A_j^{(0)} x^j, \quad (5)$$

181 where $\circ$ denotes the composition of functions, $A_j^{(l)} \in \mathbb{R}^{d_{l+1}^j \times d_l^j}$ is the weight
 182 matrix of layer l in the j -th subnetwork, and $\phi_{\mathbf{v}_l^j} : \mathbb{R}^{d_l^j} \rightarrow \mathbb{R}^{d_l^j}$ is its corre-
 183 sponding activation function associated with $\mathbf{v}_l^j = (v_{l,1}^j, \dots, v_{l,d_l^j}^j) \in \mathbb{R}^{d_l^j}$. For
 184 any vector $(a^1, \dots, a^{d_l^j}) \in \mathbb{R}^{d_l^j}$, there holds

$$\phi_{\mathbf{v}_l^j} \begin{pmatrix} a^1 \\ \vdots \\ a^{d_l^j} \end{pmatrix} = \begin{pmatrix} \max \{a^1 - v_{l,1}^j, 0\} \\ \vdots \\ \max \{a^{d_l^j} - v_{l,d_l^j}^j, 0\} \end{pmatrix}.$$

185 The function space employed by the NAMs is defined as

$$\mathcal{H} = \left\{ f : f = \sum_{j=1}^p f_j, \text{ } f_j \text{ satisfies (5)} \right\}. \quad (6)$$

186 The empirical risk, by minimizing the empirical negative log-likelihood,
 187 is denoted as

$$\mathcal{L}_n(f) = \frac{1}{n} \sum_{i=1}^n \left[-\log h(y_i - f(\mathbf{x}_i)) \right], \quad (7)$$

188 where h is the density function of the error random variable $\epsilon_i = y_i - f(\mathbf{x}_i)$
 189 (Chen et al., 2024). The population version of (7) is denoted as

$$\mathcal{L}(f) = \mathbb{E} \left[-\log h(y - f(\mathbf{x})) \right]. \quad (8)$$

190 After equipping the hypothesis space $\mathcal{H}$ in (6) and empirical risk in (7), we
 191 define the density-known estimator of ML-NAM by

$$\hat{f} = \arg \min_{f \in \mathcal{H}} \mathcal{L}_n(f). \quad (9)$$

192 Since the density function h is often unknown, we employ the kernel
 193 density estimation to derive its empirical approximation (Kanamori et al.,
 194 2012). Given kernel function $K_\sigma(u, u') = \exp(-\frac{\|u-u'\|^2}{2\sigma^2})$ with bandwidth σ
 195 and predictor f , the empirical estimator of h is

$$\hat{h}(u) = \frac{1}{n} \sum_{i=1}^n K_\sigma(y_i - f(\mathbf{x}_i), u), \quad u \in \mathbb{R}. \quad (10)$$

196 Then, the density-unknown estimator of ML-NAM is defined as

$$\tilde{f} = \arg \min_{f \in \mathcal{H}} \hat{\mathcal{L}}_n(f) = \arg \min_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \left[-\log \hat{h}(y_i - f(\mathbf{x}_i)) \right], \quad (11)$$

197 which avoids relying on prior knowledge of the density function h . From
 198 the perspective of maximum likelihood estimation, the least squares loss and
 199 the least absolute deviation loss are optimal under Gaussian and Laplacian
 200 noise assumptions, respectively. In contrast, the proposed loss function re-
 201 mains flexible and adaptable to various noise structures without requiring an
 202 assumed distribution in advance.

203 **Remark 1.** *The computational complexity of our algorithm is dominated by*
 204 *the KDE component, which exhibits $\mathcal{O}(n^2)$ complexity in its direct formula-*
 205 *tion due to pairwise kernel evaluations across all samples (Silverman, 2018).*
 206 *To mitigate this bottleneck, we employ mini-batch training as described in*
 207 *Section 4.*

208 **Remark 2.** *Furthermore, acceleration could be achieved through approxi-*
 209 *mation techniques such as random Fourier features, which reduce the kernel*
 210 *computation complexity to $\mathcal{O}(Mn)$ with $M \ll n$, where M denotes the num-*
 211 *ber of random features used to map inputs into a randomized feature space*
 212 *(Yang et al., 2003). This strategy offers a principled direction for scaling*
 213 *KDE-based methods to larger datasets in our future research.*

214 3. Learning Theory Analysis

215 This section provides our main results on the upper bounds of the excess
 216 risk of ML-NAM. The analysis proceeds by decomposing the risk into sample
 217 and approximation error terms, and derives convergence rates under mild
 218 assumptions.

219 *3.1. Notations*

220 Before presenting our main results, we introduce some relevant notations.
 221 Let $y_i = f(\mathbf{x}_i) + \epsilon_i$ with $f(\mathbf{x}_i) = \sum_{j=1}^p f_j(\mathbf{x}_i^j)$ and $\mathbb{E}(\epsilon_i) = 0$. Define

$$F(f, \mathbf{z}_i) = -\log h(y_i - f(\mathbf{x}_i)) + \log h(y_i - f^*(\mathbf{x}_i)), \quad f \in \{\hat{f}, f_{\mathcal{H}}^*\}$$

222 for any f , where f^* is defined as:

$$f^* := \arg \min_f \mathcal{L}(f) = \arg \min_f \mathbb{E}(-\log h(y - f(\mathbf{x}))). \quad (12)$$

223 Let $\mathcal{Z}'_n = \{\mathbf{z}'_i := (\mathbf{x}'_i, y'_i)\}_{i=1}^n$ be another sample independent of $\mathcal{Z}_n$, then
 224 the excess risk of $\mathcal{L}(\hat{f}) - \mathcal{L}(f^*) = \mathbb{E}_{\mathcal{Z}'_n} \left(\frac{1}{n} \sum_{i=1}^n F(\hat{f}, \mathbf{z}'_i) \right)$. Note that $\mathcal{L}(\hat{f})$ is
 225 random as $\hat{f}$ is produced based on the sample $\mathbf{z}'_i$. Naturally, we will focus on
 226 bounding the expected excess risk

$$\mathbb{E}(\mathcal{L}(\hat{f}) - \mathcal{L}(f^*)) = \mathbb{E}_{\mathcal{Z}_n} \left[\mathbb{E}_{\mathcal{Z}'_n} \left(\frac{1}{n} \sum_{i=1}^n F(\hat{f}, \mathbf{z}'_i) \right) \right].$$

227 *3.2. Error Decomposition*

228 As a stepping stone, the following error decomposition is used in our
 229 theoretical analysis.

230 Denote

$$f_{\mathcal{H}}^* := \arg \min_{f \in \mathcal{H}} \mathcal{L}(f)$$

231 as the optimal predictor in $\mathcal{H}$ with respect to the expected risk (8). For any
 232 given $\hat{f}$, we have

$$\mathbb{E}_{\mathcal{Z}'_n} \left[\sum_{i=1}^n -\log h(y'_i - f_{\mathcal{H}}^*(\mathbf{x}'_i)) \right] \leq \mathbb{E}_{\mathcal{Z}'_n} \left[\sum_{i=1}^n -\log h(y'_i - \hat{f}(\mathbf{x}'_i)) \right].$$

233 Then the excess risk can be decomposed as

$$\begin{aligned}
& \mathbb{E} \left[\mathcal{L}(\hat{f}) - \mathcal{L}(f^*) \right] \\
&= \mathbb{E}_{\mathcal{Z}_n} \left\{ \frac{1}{n} \sum_{i=1}^n \left[\mathbb{E}_{\mathcal{Z}'_n} (-\log h(y'_i - \hat{f}(\mathbf{x}'_i))) - \mathbb{E}_{\mathcal{Z}'_n} (-\log h(y'_i - f^*(\mathbf{x}'_i))) \right] \right\} \\
&\leq \mathbb{E}_{\mathcal{Z}_n} \left\{ \frac{1}{n} \sum_{i=1}^n \left[\mathbb{E}_{\mathcal{Z}'_n} \left(F(\hat{f}, \mathbf{z}'_i) \right) \right] \right\} \\
&\quad + \mathbb{E}_{\mathcal{Z}_n} \left\{ \frac{1}{n} \sum_{i=1}^n \left[\log h(y_i - \hat{f}(\mathbf{x}_i)) - \log h(y_i - f^*(\mathbf{x}_i)) \right] \right\} \\
&= \mathbb{E}_{\mathcal{Z}_n} \left\{ \frac{1}{n} \sum_{i=1}^n \left[\mathbb{E}_{\mathcal{Z}'_n} \left(F(\hat{f}, \mathbf{z}'_i) \right) - F(\hat{f}, \mathbf{z}_i) \right] \right\} + \mathbb{E}_{\mathcal{Z}_n} \left[\frac{1}{n} \sum_{i=1}^n F(f^*, \mathbf{z}_i) \right] \\
&\leq \mathbb{E}_{\mathcal{Z}_n} \left\{ \frac{1}{n} \sum_{i=1}^n \left[\mathbb{E}_{\mathcal{Z}'_n} \left(F(\hat{f}, \mathbf{z}'_i) \right) - F(\hat{f}, \mathbf{z}_i) \right] \right\} + [\mathcal{L}(f^*) - \mathcal{L}(f^*)] \\
&= I_1 + I_2,
\end{aligned} \tag{13}$$

234 where

$$I_1 = \mathbb{E}_{\mathcal{Z}_n} \left\{ \frac{1}{n} \sum_{i=1}^n \left[\mathbb{E}_{\mathcal{Z}'_n} \left(F(\hat{f}, \mathbf{z}'_i) \right) - F(\hat{f}, \mathbf{z}_i) \right] \right\}$$

235 and

$$I_2 = \left[\mathcal{L} \left(\sum_{j=1}^p f^*_{\mathcal{H}_j} \right) - \mathcal{L} \left(\sum_{j=1}^p f_j^* \right) \right].$$

236 **Remark 3.** The above decomposition lies in the adoption of double expecta-
237 tions with respect to $\mathcal{Z}_n, \mathcal{Z}'_n$ and the definition of $f^*_{\mathcal{H}}$. The error terms I_1 and
238 I_2 in (13) represent the expectations of the sample error and the approxima-
239 tion error, respectively. Note that $f^*_{\mathcal{H}}$ is a sample independent function, so I_2
240 can be further simplified to $\mathcal{L}(f^*_{\mathcal{H}}) - \mathcal{L}(f^*)$, which characterizes how well the
241 actual target function f^* can be approximated by functions in $\mathcal{H}$. The sample
242 error term I_1 quantifies the gap between the empirical risk and the expected
243 risk of the estimator $\hat{f}$.

244 3.3. Error Bounds

245 To derive the convergence rate of the excess risk, the following assump-
246 tions are used for our error analysis.

247 **Assumption 1.** (Wang et al., 2025) *Assume the density function h has a*
 248 *continuous second derivative and satisfies $h(\xi) > c > 0$ for any $\xi \in \mathbb{R}$.*

249 **Assumption 2.** (Yarotsky, 2017) *Suppose the hypothesis function space $\mathcal{H}$*
 250 *satisfies $\|f\|_\infty \leq \mathcal{B}, \forall f \in \mathcal{H}$ and the target function f^* defined in (1) also*
 251 *satisfies $\|f^*\|_\infty \leq \mathcal{B}$.*

252 The above assumptions have been widely employed in learning-theory
 253 analysis (Yarotsky, 2017; Wang et al., 2025). Assumption 1 guarantees suffi-
 254 cient smoothness of the density function. Meanwhile, Assumption 2 imposes
 255 a crucial boundedness condition on f^* and the function class $\mathcal{H}$, which is
 256 essential for deriving stable convergence rates and ensuring the analytical
 257 tractability of neural network models.

258 Before introducing Theorem 1, we present several definitions and auxiliary
 259 lemmas that will be used in the subsequent analysis.

260 **Definition 1** (Covering number, Şimşekli et al. (2021)). *A set $\{f_l\}_{l=1}^N$ is the*
 261 *ε -cover of metric space $(\mathcal{H}, d)$, if $\forall f \in \mathcal{H}$, there exists $f_k \in \{f_l\}_{l=1}^N$ such that*

$$d(f_k, f) \leq \varepsilon.$$

262 *The covering number $\mathcal{N}(\mathcal{H}, d, \varepsilon)$ is the minimum **cardinality** of all ε -covers.*

263 **Definition 2** (Pseudo-dimension, Bartlett et al. (2019)). *For a function class*
 264 *$\mathcal{H} : \mathcal{X} \rightarrow \mathbb{R}$, its pseudo-dimension $\text{Pdim}(\mathcal{H})$ is the largest integer m for which*
 265 *there exist points $(\mathbf{x}_1, \dots, \mathbf{x}_m) \in \mathcal{X}^m$ and thresholds $(y_1, \dots, y_m) \in \mathbb{R}^m$ such*
 266 *that every binary labeling $(b_1, \dots, b_m) \in \{0, 1\}^m$ is realized by some $f \in \mathcal{H}$ in*
 267 *the sense that*

$$f(\mathbf{x}_i) > y_i \iff b_i = 1, \quad i = 1, \dots, m.$$

268 The following two results, Theorem 12.2 in Anthony and Bartlett (2009)
 269 and Theorem 8 from Bartlett et al. (2019), show that the metric entropy may
 270 be bounded in terms of the pseudo-dimension.

271 **Lemma 1.** *Assume for all $f \in \mathcal{H}$, $\|f\|_\infty \leq \mathcal{B}$. Denote the pseudo-dimension*
 272 *of $\mathcal{H}$ as $\text{Pdim}(\mathcal{H})$ and $n \geq \text{Pdim}(\mathcal{H})$, we have for any ε ,*

$$\mathcal{N}(\mathcal{H}, d_\infty, \varepsilon) \leq \left(\frac{2e\mathcal{B} \cdot n}{\varepsilon \cdot \text{Pdim}(\mathcal{H})} \right)^{\text{Pdim}(\mathcal{H})}.$$

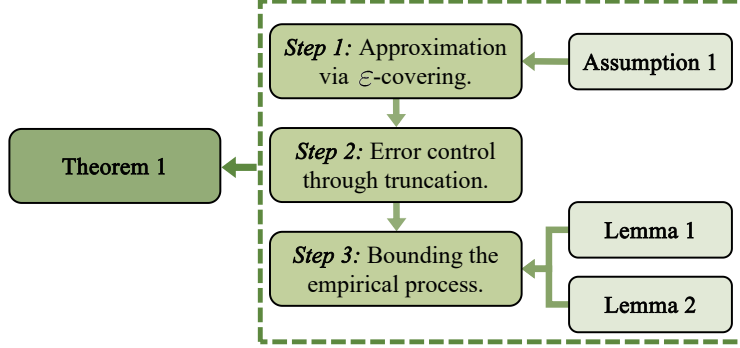


Figure 3: The proof diagram of Theorem 1.

Lemma 2. *With the notation from before, we have that*

$$\text{Pdim}(\mathcal{H}(\mathcal{S}, \mathcal{U}, \mathcal{B})) = \mathcal{O}(SU).$$

where $\mathcal{U}$ denotes the number of activation functions, and $\mathcal{S}$ denotes the total number of trainable parameters.

Now, we state our first theorem on the excess risk of ML-NAM.

Theorem 1. *Suppose that Assumptions 1-2 hold. Then, for $2n \geq \text{Pdim}(\mathcal{H})$, there exists a constant $C > 0$ satisfying*

$$\mathbb{E} \left[\mathcal{L}(\hat{f}) - \mathcal{L}(f^*) \right] \leq CB \left(\frac{\log n}{n} SU \right)^{\frac{1}{2}} + (\mathcal{L}(f_{\mathcal{H}}^*) - \mathcal{L}(f^*)).$$

Proof. To prove Theorem 1, we employ an ε -covering approximation, a truncation scheme, and the classical chaining technique for empirical processes, which together enable us to control the logarithmic loss and to establish the desired uniform convergence bound. The outline of our proof is presented in Figure 3.

Step 1: Approximation via ε -covering. We employ the ε -covering argument to approximate the continuous estimator $\hat{f}$ by its discrete surrogate f_{q^*} drawn from a finite cover. Define $D(f, \mathbf{z}_i) = \mathbb{E}_{\mathbf{z}'_i} \{F(f, \mathbf{z}'_i)\} - F(f, \mathbf{z}_i)$ for $f \in \mathcal{H}$. By the definition of $F(f, \mathbf{z}_i)$, it follows that

$$\begin{aligned} \left| D(\hat{f}, \mathbf{z}_i) - D(f_{q^*}, \mathbf{z}_i) \right| &= \left| \mathbb{E}_{\mathbf{z}'_i} \{F(\hat{f}, \mathbf{z}'_i)\} - F(\hat{f}, \mathbf{z}_i) - \mathbb{E}_{\mathbf{z}'_i} \{F(f_{q^*}, \mathbf{z}'_i)\} + F(f_{q^*}, \mathbf{z}_i) \right|, \\ \left| F(\hat{f}, \mathbf{z}_i) - F(f_{q^*}, \mathbf{z}_i) \right| &= \left| \log h(\hat{f}(\mathbf{x}_i) - y_i) - \log h(f_{q^*}(\mathbf{x}_i) - y_i) \right|. \end{aligned}$$

288 Let $\{f_j\}_{j=1}^{\mathcal{N}(\mathcal{H}, d_\infty, \varepsilon)}$ be the minimum covering of $\mathcal{H}$ with ε and metric d_∞ .
 289 There exists $f_{q^*} \in \{f_j\}_{j=1}^{\mathcal{N}(\mathcal{H}, d_\infty, \varepsilon)}$ such that $\max_{1 \leq i \leq n} |\hat{f}(\mathbf{x}_i) - f_{q^*}(\mathbf{x}_i)| \leq \varepsilon$.
 290 Then, we have

$$\begin{aligned}
 & \left| \frac{1}{n} \sum_{i=1}^n \mathbb{E}_{\mathcal{Z}_n} \{F(\hat{f}, \mathbf{z}_i)\} - \frac{1}{n} \sum_{i=1}^n \mathbb{E}_{\mathcal{Z}_n} \{F(f_{q^*}, \mathbf{z}_i)\} \right| \\
 & \leq \frac{1}{n} \sum_{i=1}^n \mathbb{E}_{\mathcal{Z}_n} |F(\hat{f}, \mathbf{z}_i) - F(f_{q^*}, \mathbf{z}_i)| \\
 & = \frac{1}{n} \sum_{i=1}^n \mathbb{E}_{\mathcal{Z}_n} |\log h(\hat{f}(\mathbf{x}_i) - y_i) - \log h(f_{q^*}(\mathbf{x}_i) - y_i)| \\
 & = \frac{1}{n} \sum_{i=1}^n \mathbb{E}_{\mathcal{Z}_n} \left| \frac{h^{(1)}(\xi)(\hat{f}(\mathbf{x}_i) - f_{q^*}(\mathbf{x}_i))}{h(\xi)} \right| \\
 & \leq \mathcal{B}\varepsilon,
 \end{aligned}$$

291 where the last inequality follows from Assumption 1, and

$$\begin{aligned}
 & \mathbb{E}_{\mathcal{Z}_n} \left\{ \frac{1}{n} \sum_{i=1}^n \left| D(\hat{f}, \mathbf{z}_i) - D(f_{q^*}, \mathbf{z}_i) \right| \right\} \\
 & = \mathbb{E}_{\mathcal{Z}_n} \left\{ \frac{1}{n} \sum_{i=1}^n \left| \mathbb{E}_{\mathcal{Z}'_n} \{F(\hat{f}, \mathbf{z}'_i)\} - F(\hat{f}, \mathbf{z}_i) - \mathbb{E}_{\mathcal{Z}'_n} \{F(f_{q^*}, \mathbf{z}'_i)\} + F(f_{q^*}, \mathbf{z}_i) \right| \right\} \\
 & \leq \mathbb{E}_{\mathcal{Z}_n} \left\{ \frac{1}{n} \sum_{i=1}^n \left| \mathbb{E}_{\mathcal{Z}'_n} \{F(\hat{f}, \mathbf{z}'_i)\} - \mathbb{E}_{\mathcal{Z}'_n} \{F(f_{q^*}, \mathbf{z}'_i)\} \right| + \left| F(\hat{f}, \mathbf{z}_i) - F(f_{q^*}, \mathbf{z}_i) \right| \right\} \\
 & \leq 2\mathcal{B}\varepsilon.
 \end{aligned} \tag{14}$$

292 *Step 2: Error control through truncation.* This step bounds the deviation
 293 between the original unbounded logarithmic loss and its truncated version,
 294 ensuring that the truncation bias is controlled under sufficient tail decay. Let
 295 $\alpha_n \geq 0$ be a threshold that depends on the sample size n . Denote $h_T = h$ if
 296 $h \geq \alpha_n$ and $h_T = \alpha_n$ otherwise. Define

$$f_{\alpha_n}^*(x) = \arg \min_{\|f\|_\infty \leq \mathcal{B}} \mathbb{E}(-\log h_T(y - f(\mathbf{x})) \mid \mathbf{x} = x),$$

297 and

$$F_{\alpha_n}(f, \mathbf{z}_i) = -\log h_T(y_i - f(\mathbf{x}_i)) + \log h_T(y_i - f_{\alpha_n}^*(\mathbf{x}_i)), \quad f \in \mathcal{H}.$$

298 Then, we have

$$\begin{aligned}
\mathbb{E} \{F(f, \mathbf{z}_i)\} &= \mathbb{E} \{F_{\alpha_n}(f, \mathbf{z}_i)\} + \mathbb{E} \{\log h_T(y_i - f(\mathbf{x}_i)) - \log h(y_i - f(\mathbf{x}_i))\} \\
&\quad + \mathbb{E} \{\log h_T(y_i - f^*(\mathbf{x}_i)) - \log h_T(y_i - f_{\alpha_n}^*(\mathbf{x}_i))\} \\
&\quad + \mathbb{E} \{\log h(y_i - f^*(\mathbf{x}_i)) - \log h_T(y_i - f^*(\mathbf{x}_i))\} \\
&\leq \mathbb{E} \{F_{\alpha_n}(f, \mathbf{z}_i)\} + \mathbb{E} |\log h_T(y_i - f(\mathbf{x}_i)) - \log h(y_i - f(\mathbf{x}_i))| \\
&\quad + \mathbb{E} |\log h(y_i - f^*(\mathbf{x}_i)) - \log h_T(y_i - f^*(\mathbf{x}_i))| \\
&\leq \mathbb{E} \{F_{\alpha_n}(f, \mathbf{z}_i)\} + 2\mathbb{E} |(\log \alpha_n - \log f(\xi))\mathbb{I}(f(\xi) < \alpha_n)| \\
&\leq \mathbb{E} \{F_{\alpha_n}(f, \mathbf{z}_i)\} + 4\mathbb{E} \{|\log f(\xi)\mathbb{I}(f(\xi) < \alpha_n)|\}.
\end{aligned}$$

299 Similarly, we can divide

$$\begin{aligned}
\mathbb{E} \{F_{\alpha_n}(f, \mathbf{z}_i)\} &= \mathbb{E} \{F(f, \mathbf{z}_i)\} + \mathbb{E} \{\log h(y_i - f(\mathbf{x}_i)) - \log h_T(y_i - f(\mathbf{x}_i))\} \\
&\quad + \mathbb{E} \{\log h(y_i - f_{\alpha_n}^*(\mathbf{x}_i)) - \log h(y_i - f^*(\mathbf{x}_i))\} \\
&\quad + \mathbb{E} \{\log h_T(y_i - f_{\alpha_n}^*(\mathbf{x}_i)) - \log h(y_i - f_{\alpha_n}^*(\mathbf{x}_i))\} \\
&\leq \mathbb{E} \{F(f, \mathbf{z}_i)\} + \mathbb{E} |\log h_T(y_i - f(\mathbf{x}_i)) - \log h(y_i - f(\mathbf{x}_i))| \\
&\quad + \mathbb{E} |\log h(y_i - f_{\alpha_n}^*(\mathbf{x}_i)) - \log h_T(y_i - f_{\alpha_n}^*(\mathbf{x}_i))| \\
&\leq \mathbb{E} \{F(f, \mathbf{z}_i)\} + 4\mathbb{E} \{|\log f(\xi)\mathbb{I}(f(\xi) < \alpha_n)|\}.
\end{aligned}$$

300 Combining the above two inequalities, we have

$$\left| \frac{1}{n} \sum_{i=1}^n \mathbb{E} \{F(f, \mathbf{z}_i)\} - \frac{1}{n} \sum_{i=1}^n \mathbb{E} \{F_{\alpha_n}(f, \mathbf{z}_i)\} \right| \leq 4\mathbb{E} \{|\log f(\xi)\mathbb{I}(f(\xi) < \alpha_n)|\}.$$

301 Furthermore, we can get

$$\left| \mathbb{E}_{Z_n} \left[\frac{1}{n} \sum_{i=1}^n (D(f_{q^*}, \mathbf{z}_i) - D_{\alpha_n}(f_{q^*}, \mathbf{z}_i)) \right] \right| \leq 8\mathbb{E} \{|\log f(\xi)\mathbb{I}(f(\xi) < \alpha_n)|\}. \tag{15}$$

302 *Step 3: Bounding the empirical process.* Finally, we apply Dudley's chain-
303 ing together with concentration inequalities to control the stochastic fluctu-

304 ations. Consequently, we have

$$\begin{aligned}
\mathbb{E}_{\mathcal{Z}_n} \left[\frac{1}{n} \sum_{i=1}^n D_{\alpha_n}(f_{q^*}, \mathbf{z}_i) \right] &= \mathbb{E}_{\mathcal{Z}_n} \left\{ \mathbb{E}_{\mathcal{Z}'_n} [\zeta_i F_{\alpha_n}(f_{q^*}, \mathbf{z}'_i)] - \frac{1}{n} \sum_{i=1}^n \zeta_i F_{\alpha_n}(f_{q^*}, \mathbf{z}_i) \right\} \\
&\leq 4 \mathbb{E}_{\mathcal{Z}_n} \left\{ \sup_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \zeta_i F_{\alpha_n}(f_{q^*}, \mathbf{z}_i) \right\} \\
&\leq 4 \mathcal{B} \mathbb{E}_{\mathcal{Z}_n} \left\{ \sup_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \zeta_i \frac{F_{\alpha_n}(f_{q^*}, \mathbf{z}_i)}{\mathcal{B}} \right\} \\
&\leq \frac{C\mathcal{B}}{\sqrt{n}} \int_0^2 \sqrt{\log \mathcal{N}(\mathcal{H}/\mathcal{B}, d_\infty, \varepsilon)} d\varepsilon \\
&\leq \frac{C\mathcal{B}}{\sqrt{n}} \int_0^2 \sqrt{\log \left(\left(\frac{2e \cdot n}{\varepsilon \cdot \text{Pdim}(\mathcal{H})} \right)^{\text{Pdim}(\mathcal{H})} \right)} d\varepsilon \\
&\leq C\mathcal{B} \left(\frac{\mathcal{S} \cdot \mathcal{U} \log n}{n} \right)^{\frac{1}{2}}
\end{aligned} \tag{16}$$

305 for some constant $C > 0$. Here, the first inequality follows from the sym-
306 metrization and Talagrand's concentration inequality (Talagrand, 2014), the
307 third follows from Dudley's Chaining, the fourth from Lemma 1, and the
308 last inequality from Lemma 2. Combining inequalities (14), (15) and (16),
309 we have

$$\begin{aligned}
I_1 &= \mathbb{E}_{\mathcal{Z}_n} \left\{ \frac{1}{n} \sum_{i=1}^n D(\hat{f}, \mathbf{z}_i) \right\} \\
&\leq \mathbb{E}_{\mathcal{Z}_n} \left\{ \frac{1}{n} \sum_{i=1}^n D(f_{q^*}, \mathbf{z}_i) \right\} + 2\mathcal{B}\varepsilon \\
&\leq \mathbb{E}_{\mathcal{Z}_n} \left\{ \frac{1}{n} \sum_{i=1}^n D_{\alpha_n}(f_{q^*}, \mathbf{z}_i) \right\} + 8\mathbb{E}\{ |\log f(\xi) \mathbb{I}(f(\xi) < \alpha_n)| \} + 2\mathcal{B}\varepsilon \\
&\leq C\mathcal{B} \left(\frac{\mathcal{S} \cdot \mathcal{U} \log n}{n} \right)^{\frac{1}{2}} + 8\mathbb{E}\{ |\log f(\xi) \mathbb{I}(f(\xi) < \alpha_n)| \} + 2\mathcal{B}\varepsilon.
\end{aligned}$$

310 This completes the proof. $\square$

311 **Remark 4.** To derive the upper bound in Theorem 1, we control the cover-
312 ing number of $\mathcal{H}$ via its pseudo-dimension. Following Bartlett et al. (2019),

the pseudo-dimension of $\mathcal{H}$, when using piecewise polynomial activation functions, can be bounded in terms of the total number of parameters $\mathcal{S}$ and the number of nonlinear units $\mathcal{U}$, yielding $\text{Pdim}(\mathcal{H}) = \mathcal{O}(\mathcal{S}\mathcal{U})$. This result yields an upper bound on the prediction error, which is the sum of the sample error and the approximation error between $\mathcal{H}$ and the ground-truth function f^* .

Remark 5. Theorem 1 demonstrates that the excess risk of ML-NAM depends on the sample size n , the network parameters, and the approximation error. For instance, when $\mathcal{S}$ and $\mathcal{U}$ are constant with respect to n , then the rate becomes $n^{-1/2} + I_2$.

Furthermore, we investigate the theoretical properties of the ML-NAM estimator $\hat{f}$ in (9) associated with Besov spaces. The Besov space is a considerably general function space including the Hölder space and the Sobolev space, and especially can capture the spatial inhomogeneity of smoothness.

Definition 3. (Sawano et al., 2018) For a measurable domain Ω and $p \in (0, \infty]$, denote $L^p(\Omega) = \left\{ f: \Omega \rightarrow \mathbb{R} \mid \|f\|_p < \infty \right\}$ with $\|f\|_p = \left(\int_{\Omega} |f(\mathbf{x})|^p d\mathbf{x} \right)^{1/p}$. For any function $f \in L^p(\Omega)$, the r -th modulus of smoothness is defined as

$$w_{r,q}(f, t) = \sup_{\|\mathbf{u}\|_2 \leq t} \|\Delta_{\mathbf{u}}^r(f)\|_q, \quad \mathbf{u} \in \Omega,$$

where $\Delta_{\mathbf{u}}^r(f)(\mathbf{x}) = \sum_{j=0}^r \binom{r}{j} (-1)^{r-j} f(\mathbf{x} + \mathbf{u}j)$.

Definition 4. (Sawano et al., 2018) For $0 < q, k \leq \infty, s > 0, r := \lfloor s \rfloor + 1$, we define the Besov space $B_{q,k}^s(\Omega)$ as

$$B_{q,k}^s(\Omega) = \{f \in L^q(\Omega) \mid \|f\|_{B_{q,k}^s} < \infty\},$$

where $\|f\|_{B_{q,k}^s} := \|f\|_q + |f|_{B_{q,k}^s}$ with

$$|f|_{B_{q,k}^s} := \begin{cases} \left(\int_0^\infty (t^{-s} w_{r,q}(f, t))^k \frac{dt}{t} \right)^{\frac{1}{k}} & \text{if } k < \infty, \\ \sup_{t>0} t^{-s} w_{r,q}(f, t) & \text{otherwise.} \end{cases}$$

Before stating the excess risk bound for $\hat{f}$, we introduce the assumption for the target function f^* in (1).

Assumption 3 (Padilla et al. (2022)). The target function $f^* = \sum_{j=1}^p f_j^*$, $f_j^* \in B_{q,k}^{s_j}([0, 1])$ and $\|f_j^*\|_\infty \leq \mathcal{B}$, where $0 < q, k \leq \infty, 0 < s_j < \infty$ with $s_j \geq 1/q$. Furthermore, there exists $m_j \in \mathbb{N}$ such that $0 < s_j < \min\{m_j, m_j - 1 + 1/q\}$.

339 Assumption 3 imposes the necessary smoothness condition on the target
 340 function f^* . We are now ready to state the main result on the estimation of
 341 f^* when f^* belongs to a Besov space.

342 **Theorem 2.** *Suppose that Assumptions 1-3 hold. For the j -th subnetwork,*
 343 *denote $B_j = \max_{1 \leq l \leq L_j} (\|A_j^{(l)}\|_\infty + \|\mathbf{v}_l^j\|_\infty)$, $W_j = \max_{1 \leq l \leq L_j+1} d_l^j$, and its*
 344 *number of non-zero parameters as S_j . Then, as $n \rightarrow \infty$, there exists a*
 345 *constant $C' > 0$ such that*

$$\mathbb{E} \left[\mathcal{L}(\hat{f}) - \mathcal{L}(f^*) \right] \leq C' \mathcal{B} \left\{ \left(\frac{S \mathcal{U} \log n}{n} \right)^{\frac{1}{2}} + p \max_{j=1, \dots, p} N_j^{-s_j} \right\},$$

346 *when choosing the parameters of each subnetwork class as*

$$L_j = 3 + 2 \lceil \log_2 \left(\frac{3^{\max(1, m_j)}}{\delta_j c_{1, m_j}} \right) + 5 \rceil \lceil \log_2 \max(1, m_j) \rceil, \quad W_j = W_0 N_j,$$

$$S_j = (L_j - 1) W_0^2 N_j + N_j, \quad B_j = \mathcal{O} \left(N_j^{(v_j^{-1} + 1)(\max\{1, (1/q - s_j)_+\})} \right),$$

347 *for a constant $c_{1, m_j} > 0$ that depends on $m_j, j = 1, \dots, p$, a constant $W_0 > 0$,*

348 *where $v_j = q s_j - 1, \delta_j = N_j^{-s_j - (v_j^{-1} + 1)(1/q - s_j)} + \{\log N_j\}^{-1}, N_j \asymp n^{\frac{1}{2s_j + 1}}$.*

349 *Proof.* Form the definition of $\mathcal{L}(f)$ in (8), we have

$$\begin{aligned} \mathcal{L}(f_{\mathcal{H}}^*) - \mathcal{L}(f^*) &= \mathbb{E}_{Z'_n} [-\log h(y - f_{\mathcal{H}}^*(\mathbf{x})) + \log h(y - f^*(\mathbf{x}))] \\ &\leq \mathbb{E}_{Z'_n} \left| \frac{h^{(1)}(\xi)}{h(\xi)} (f_{\mathcal{H}}^*(\mathbf{x}) - f^*(\mathbf{x})) \right| \\ &\leq \mathcal{B} \mathbb{E}_{Z'_n} |f_{\mathcal{H}}^*(\mathbf{x}) - f^*(\mathbf{x})| \\ &\leq \mathcal{B} \sum_{j=1}^p \mathbb{E}_{Z'_n} |f_{\mathcal{H}_j}^*(\mathbf{x}) - f_j^*(\mathbf{x})|. \end{aligned}$$

350 Then, according to Proposition 1 in Suzuki (2019), let $f^* \in B_{q, k}^s, f_{\mathcal{H}}^* \in \mathcal{H}$,

351 the worst case approximation error for a p -dimensional function satisfies

$$\sup_{f^*} \inf_{f_{\mathcal{H}}^*} \|f^* - f_{\mathcal{H}}^*\|_{\ell^r([0, 1]^p)} \lesssim N^{-s/p}.$$

352 In the additive setting, we can get $|f_{\mathcal{H}_j}^*(\mathbf{x}) - f_j^*(\mathbf{x})| \leq c_1 N_j^{-s_j}$, and then

$$\mathcal{L}(f_{\mathcal{H}}^*) - \mathcal{L}(f^*) \leq \mathcal{B} c_1 p \max_{j=1, \dots, p} N_j^{-s_j}.$$

353 Combining the results of I_1 and I_2 , we obtain the desired result. $\square$

Remark 6. Theorem 2 indicates that the smoothness s_j of each component f_j determines its approximation complexity (Suzuki, 2019). Components with lower smoothness thus require more parameters to achieve a comparable level of approximation accuracy. This motivates a heterogeneous network design (Tadipatri et al., 2025). While all subnetworks maintain a depth L_j of logarithmic order in $1/\delta_j$, their width W_j and sparsity S_j depend on the corresponding smoothness level. Components with low smoothness are assigned wider subnetworks, whereas smooth components can be represented compactly. Such adaptive capacity allocation enables ML-NAM to utilize componentwise smoothness, outperforming linear models that assume a uniform smoothness level.

Remark 7. The auxiliary parameters (e.g., q, m_j, v_j, δ_j) have only limited influence on the dominant asymptotic convergence rate $N_j^{-s_j}$. Instead, they characterize the admissible function class and the architectural conditions under which this rate is attainable, thereby affecting only the corresponding constant factors (Bolcskei et al., 2019). The decay of the approximation error is essentially governed by the sample size N_j and the smoothness level s_j .

Combining this result with Theorem 2 in Suzuki (2019), and utilizing the equilibrium between sample error I_1 and approximation error I_2 established therein, we derive the optimal convergence rate, as stated in the following corollary.

Corollary 1. Under the conditions in Theorem 2, there exists a constant $C_1 > 0$ such that

$$\mathbb{E} \left[\mathcal{L}(\hat{f}) - \mathcal{L}(f^*) \right] \leq C_1 n^{-\frac{2 \min_j s_j}{2 \min_j s_j + 1}} (\log n)^2,$$

with probability approaching one.

Remark 8. Corollary 1 shows that $\hat{f}$ attains a polynomially decaying rate in estimating f^* under much more general noise conditions, which is comparable to previous analyses in Christmann and Zhou (2016); Chen et al. (2020). In particular, since the term with the smallest smoothness index dictates the asymptotic behavior of the estimation error, the resulting convergence rate is consequently governed by $\min_j s_j$.

When the density function is unknown, an additional divergence term arises from substituting h with its kernel estimator $\hat{h}$. Based on Theorem 2 in Wang et al. (2025), we obtain the following corollary.

387 **Corollary 2.** *If h is unknown, assume for any given $\sigma > 0$, $n\sigma \rightarrow \infty$ as*
 388 *$n \rightarrow \infty$. For $\tilde{f}$ defined in (11), under Assumptions 1–3, there exist positive*
 389 *constants C', C_2 satisfying*

$$\mathbb{E} \left[\mathcal{L}(\tilde{f}) - \mathcal{L}(f^*) \right] \leq C' \mathcal{B} \left\{ \left(\frac{SU \log n}{n} \right)^{\frac{1}{2}} + p \max_{j=1, \dots, p} N_j^{-s_j} \right\} + C_2 \sigma^2.$$

390 **Remark 9.** *In Corollary 2, σ^2 describes the error arising from approximat-*
 391 *ing h with its kernel estimator $\hat{h}$. While this substitution does not increase*
 392 *the variance, it can introduce notable bias, particularly for larger values of*
 393 *σ . This highlights the importance of choosing a smaller σ to control the*
 394 *approximation bias.*

395 4. Numerical Experiments

396 This section presents the numerical experiments conducted to evaluate
 397 ML-NAM on both simulated and real-world datasets. All experiments are
 398 conducted in Python on a workstation with an Intel(R) Xeon(R) Platinum
 399 8175M CPU and an NVIDIA RTX A6000 GPU.

400 **Baselines.** We compare ML-NAM with several representative baselines:
 401 MLP, XGBoost, Neural Additive Models (NAM, Agarwal et al. (2021)), Neu-
 402 ral Basis Models (NBM, Radenovic et al. (2022)), and Concept-based Taylor
 403 Additive Models (CAT, Duong et al. (2024)). To ensure fairness, each exper-
 404 iment is repeated 100 times. We report both the average MSE and average
 405 mean absolute error (MAE), along with their corresponding standard devia-
 406 tions.

407 **Parameter Selection.** Hyperparameters of all baseline models are
 408 tuned using a unified procedure. For the subnetwork architecture, we con-
 409 sider both MLP and spline variants. In the case of MLP subnetworks, the
 410 number of hidden layers is selected from $\{1, 2, 3\}$ and the number of units
 411 per layer from $\{16, 32, 64\}$; for spline subnetworks, the number of knots is
 412 tuned from $\{20, 25\}$. The learning rate is initialized at 10^{-3} and decayed
 413 according to a polynomial schedule (power = 0.5). Batch size is chosen from
 414 $\{32, 64, 128\}$, kernel bandwidth σ from $\{0.01, 0.05, 0.1, 0.5\}$, and dropout rate
 415 from $\{0, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5\}$. For ML-NAM⁺, an additional feature se-
 416 lection regularization term is applied, with its coefficient tuned from $\{10^{-4},$
 417 $10^{-3}, 10^{-2}, 10^{-1}\}$. Other parameters follow the settings in the original publi-
 418 cations. The optimal configuration is determined via 5-fold cross-validation.

Table 3: Experimental results on the simulation datasets, where the upper and lower panels report the average MSE ($\downarrow$) and average MAE ($\downarrow$) with their standard deviations, respectively. The best results among all baselines are highlighted in **bold**.

Outliers	Noise	(n, p)	MLP	XGBoost	NAM	NBM	CAT	ML-NAM	ML-NAM ⁺
$r = 0\%$	Normal Gaussian	(1024,10)	0.056 $\pm$ 0.012	0.029 $\pm$ 0.011	0.021 $\pm$ 0.013	0.014 $\pm$ 0.007	0.030 $\pm$ 0.007	0.014 $\pm$ 0.005	0.011 $\pm$ 0.002
		(1024,100)	0.123 $\pm$ 0.036	0.103 $\pm$ 0.021	0.120 $\pm$ 0.031	0.119 $\pm$ 0.034	0.115 $\pm$ 0.029	0.110 $\pm$ 0.025	0.112 $\pm$ 0.023
		(1024,200)	0.154 $\pm$ 0.057	0.139 $\pm$ 0.052	0.137 $\pm$ 0.051	0.120 $\pm$ 0.053	0.193 $\pm$ 0.048	0.125 $\pm$ 0.038	0.118 $\pm$ 0.034
		(1024,400)	0.333 $\pm$ 0.109	0.280 $\pm$ 0.096	0.266 $\pm$ 0.106	0.284 $\pm$ 0.103	0.271 $\pm$ 0.098	0.167 $\pm$ 0.072	0.154 $\pm$ 0.064
	Gaussian Mixture	(1024,10)	0.065 $\pm$ 0.035	0.079 $\pm$ 0.027	0.071 $\pm$ 0.024	0.041 $\pm$ 0.021	0.069 $\pm$ 0.022	0.035 $\pm$ 0.015	0.043 $\pm$ 0.019
		(1024,100)	0.248 $\pm$ 0.078	0.230 $\pm$ 0.071	0.219 $\pm$ 0.067	0.221 $\pm$ 0.062	0.197 $\pm$ 0.041	0.156 $\pm$ 0.036	0.137 $\pm$ 0.033
		(1024,200)	0.286 $\pm$ 0.127	0.253 $\pm$ 0.121	0.276 $\pm$ 0.113	0.241 $\pm$ 0.118	0.283 $\pm$ 0.115	0.186 $\pm$ 0.097	0.172 $\pm$ 0.094
		(1024,400)	0.673 $\pm$ 0.178	0.449 $\pm$ 0.168	0.502 $\pm$ 0.171	0.519 $\pm$ 0.174	0.495 $\pm$ 0.163	0.353 $\pm$ 0.141	0.339 $\pm$ 0.139
	Student-t	(1024,10)	0.061 $\pm$ 0.051	0.049 $\pm$ 0.038	0.059 $\pm$ 0.046	0.049 $\pm$ 0.042	0.063 $\pm$ 0.052	0.032 $\pm$ 0.024	0.023 $\pm$ 0.017
		(1024,100)	0.181 $\pm$ 0.086	0.155 $\pm$ 0.083	0.169 $\pm$ 0.087	0.178 $\pm$ 0.082	0.144 $\pm$ 0.099	0.142 $\pm$ 0.057	0.139 $\pm$ 0.052
		(1024,200)	0.302 $\pm$ 0.134	0.254 $\pm$ 0.127	0.218 $\pm$ 0.129	0.215 $\pm$ 0.121	0.201 $\pm$ 0.123	0.157 $\pm$ 0.106	0.154 $\pm$ 0.101
		(1024,400)	0.354 $\pm$ 0.165	0.289 $\pm$ 0.153	0.283 $\pm$ 0.152	0.301 $\pm$ 0.154	0.271 $\pm$ 0.152	0.191 $\pm$ 0.119	0.187 $\pm$ 0.114
$r = 10\%$	Normal Gaussian	(1024,10)	0.085 $\pm$ 0.028	0.051 $\pm$ 0.023	0.045 $\pm$ 0.034	0.042 $\pm$ 0.022	0.052 $\pm$ 0.047	0.045 $\pm$ 0.016	0.033 $\pm$ 0.011
		(1024,100)	0.151 $\pm$ 0.051	0.122 $\pm$ 0.048	0.124 $\pm$ 0.046	0.126 $\pm$ 0.041	0.121 $\pm$ 0.043	0.120 $\pm$ 0.038	0.117 $\pm$ 0.032
		(1024,200)	0.203 $\pm$ 0.106	0.159 $\pm$ 0.102	0.148 $\pm$ 0.097	0.146 $\pm$ 0.092	0.212 $\pm$ 0.085	0.147 $\pm$ 0.057	0.132 $\pm$ 0.053
		(1024,400)	0.345 $\pm$ 0.124	0.283 $\pm$ 0.112	0.335 $\pm$ 0.119	0.286 $\pm$ 0.117	0.281 $\pm$ 0.116	0.235 $\pm$ 0.086	0.207 $\pm$ 0.077
	Gaussian Mixture	(1024,10)	0.088 $\pm$ 0.058	0.113 $\pm$ 0.052	0.092 $\pm$ 0.048	0.074 $\pm$ 0.046	0.072 $\pm$ 0.043	0.069 $\pm$ 0.028	0.065 $\pm$ 0.023
		(1024,100)	0.267 $\pm$ 0.101	0.252 $\pm$ 0.103	0.256 $\pm$ 0.094	0.243 $\pm$ 0.085	0.217 $\pm$ 0.097	0.208 $\pm$ 0.062	0.198 $\pm$ 0.058
		(1024,200)	0.355 $\pm$ 0.156	0.343 $\pm$ 0.151	0.324 $\pm$ 0.152	0.312 $\pm$ 0.149	0.323 $\pm$ 0.142	0.315 $\pm$ 0.108	0.309 $\pm$ 0.101
		(1024,400)	0.713 $\pm$ 0.185	0.511 $\pm$ 0.176	0.509 $\pm$ 0.172	0.525 $\pm$ 0.178	0.523 $\pm$ 0.171	0.438 $\pm$ 0.125	0.425 $\pm$ 0.121
	Student-t	(1024,10)	0.086 $\pm$ 0.057	0.078 $\pm$ 0.047	0.068 $\pm$ 0.052	0.051 $\pm$ 0.046	0.068 $\pm$ 0.054	0.048 $\pm$ 0.037	0.042 $\pm$ 0.032
		(1024,100)	0.227 $\pm$ 0.108	0.209 $\pm$ 0.091	0.219 $\pm$ 0.093	0.201 $\pm$ 0.102	0.176 $\pm$ 0.107	0.155 $\pm$ 0.086	0.145 $\pm$ 0.081
		(1024,200)	0.359 $\pm$ 0.158	0.267 $\pm$ 0.156	0.256 $\pm$ 0.148	0.236 $\pm$ 0.145	0.215 $\pm$ 0.147	0.172 $\pm$ 0.122	0.168 $\pm$ 0.119
		(1024,400)	0.415 $\pm$ 0.186	0.342 $\pm$ 0.189	0.351 $\pm$ 0.165	0.318 $\pm$ 0.158	0.286 $\pm$ 0.154	0.248 $\pm$ 0.138	0.236 $\pm$ 0.135
Outliers	Noise	(n, p)	MLP	XGBoost	NAM	NBM	CAT	ML-NAM	ML-NAM ⁺
$r = 0\%$	Normal Gaussian	(1024,10)	0.187 $\pm$ 0.008	0.127 $\pm$ 0.006	0.108 $\pm$ 0.010	0.103 $\pm$ 0.005	0.122 $\pm$ 0.006	0.078 $\pm$ 0.004	0.071 $\pm$ 0.003
		(1024,100)	0.212 $\pm$ 0.022	0.146 $\pm$ 0.014	0.194 $\pm$ 0.021	0.183 $\pm$ 0.019	0.181 $\pm$ 0.016	0.151 $\pm$ 0.019	0.148 $\pm$ 0.018
		(1024,200)	0.318 $\pm$ 0.054	0.292 $\pm$ 0.048	0.252 $\pm$ 0.041	0.243 $\pm$ 0.042	0.325 $\pm$ 0.047	0.223 $\pm$ 0.035	0.208 $\pm$ 0.031
		(1024,400)	0.439 $\pm$ 0.106	0.413 $\pm$ 0.095	0.336 $\pm$ 0.090	0.352 $\pm$ 0.094	0.346 $\pm$ 0.093	0.287 $\pm$ 0.081	0.265 $\pm$ 0.073
	Gaussian Mixture	(1024,10)	0.211 $\pm$ 0.028	0.229 $\pm$ 0.023	0.217 $\pm$ 0.019	0.151 $\pm$ 0.016	0.204 $\pm$ 0.017	0.145 $\pm$ 0.012	0.147 $\pm$ 0.015
		(1024,100)	0.362 $\pm$ 0.064	0.353 $\pm$ 0.060	0.313 $\pm$ 0.057	0.333 $\pm$ 0.058	0.306 $\pm$ 0.052	0.264 $\pm$ 0.031	0.250 $\pm$ 0.029
		(1024,200)	0.438 $\pm$ 0.112	0.368 $\pm$ 0.113	0.373 $\pm$ 0.118	0.349 $\pm$ 0.115	0.436 $\pm$ 0.117	0.322 $\pm$ 0.103	0.287 $\pm$ 0.084
		(1024,400)	0.587 $\pm$ 0.172	0.472 $\pm$ 0.164	0.508 $\pm$ 0.167	0.526 $\pm$ 0.161	0.491 $\pm$ 0.166	0.434 $\pm$ 0.126	0.409 $\pm$ 0.115
	Student-t	(1024,10)	0.184 $\pm$ 0.042	0.156 $\pm$ 0.038	0.177 $\pm$ 0.039	0.183 $\pm$ 0.035	0.188 $\pm$ 0.036	0.127 $\pm$ 0.034	0.112 $\pm$ 0.029
		(1024,100)	0.313 $\pm$ 0.075	0.256 $\pm$ 0.064	0.261 $\pm$ 0.066	0.253 $\pm$ 0.067	0.254 $\pm$ 0.064	0.224 $\pm$ 0.053	0.206 $\pm$ 0.048
		(1024,200)	0.462 $\pm$ 0.114	0.394 $\pm$ 0.109	0.370 $\pm$ 0.110	0.364 $\pm$ 0.111	0.356 $\pm$ 0.115	0.310 $\pm$ 0.095	0.284 $\pm$ 0.093
		(1024,400)	0.497 $\pm$ 0.125	0.454 $\pm$ 0.124	0.441 $\pm$ 0.126	0.442 $\pm$ 0.124	0.423 $\pm$ 0.128	0.397 $\pm$ 0.116	0.377 $\pm$ 0.108
$r = 10\%$	Normal Gaussian	(1024,10)	0.226 $\pm$ 0.027	0.182 $\pm$ 0.022	0.158 $\pm$ 0.024	0.152 $\pm$ 0.021	0.183 $\pm$ 0.022	0.121 $\pm$ 0.013	0.118 $\pm$ 0.009
		(1024,100)	0.292 $\pm$ 0.060	0.254 $\pm$ 0.054	0.282 $\pm$ 0.039	0.272 $\pm$ 0.045	0.273 $\pm$ 0.042	0.226 $\pm$ 0.036	0.211 $\pm$ 0.028
		(1024,200)	0.363 $\pm$ 0.113	0.318 $\pm$ 0.110	0.306 $\pm$ 0.112	0.298 $\pm$ 0.108	0.377 $\pm$ 0.111	0.253 $\pm$ 0.055	0.229 $\pm$ 0.051
		(1024,400)	0.474 $\pm$ 0.121	0.434 $\pm$ 0.116	0.453 $\pm$ 0.118	0.431 $\pm$ 0.115	0.417 $\pm$ 0.116	0.382 $\pm$ 0.087	0.372 $\pm$ 0.082
	Gaussian Mixture	(1024,10)	0.246 $\pm$ 0.047	0.263 $\pm$ 0.025	0.249 $\pm$ 0.043	0.218 $\pm$ 0.042	0.207 $\pm$ 0.044	0.155 $\pm$ 0.026	0.149 $\pm$ 0.021
		(1024,100)	0.394 $\pm$ 0.119	0.399 $\pm$ 0.114	0.391 $\pm$ 0.112	0.381 $\pm$ 0.113	0.368 $\pm$ 0.114	0.282 $\pm$ 0.075	0.255 $\pm$ 0.064
		(1024,200)	0.469 $\pm$ 0.147	0.474 $\pm$ 0.133	0.441 $\pm$ 0.135	0.437 $\pm$ 0.132	0.440 $\pm$ 0.134	0.333 $\pm$ 0.099	0.299 $\pm$ 0.093
		(1024,400)	0.657 $\pm$ 0.188	0.579 $\pm$ 0.168	0.576 $\pm$ 0.164	0.582 $\pm$ 0.172	0.581 $\pm$ 0.179	0.536 $\pm$ 0.115	0.486 $\pm$ 0.113
	Student-t	(1024,10)	0.241 $\pm$ 0.051	0.222 $\pm$ 0.043	0.236 $\pm$ 0.045	0.219 $\pm$ 0.044	0.224 $\pm$ 0.046	0.131 $\pm$ 0.042	0.115 $\pm$ 0.039
		(1024,100)	0.371 $\pm$ 0.096	0.352 $\pm$ 0.072	0.369 $\pm$ 0.076	0.353 $\pm$ 0.081	0.346 $\pm$ 0.070	0.253 $\pm$ 0.072	0.216 $\pm$ 0.068
		(1024,200)	0.471 $\pm$ 0.130	0.397 $\pm$ 0.128	0.411 $\pm$ 0.121	0.384 $\pm$ 0.123	0.368 $\pm$ 0.126	0.318 $\pm$ 0.109	0.298 $\pm$ 0.107
		(1024,400)	0.549 $\pm$ 0.169	0.471 $\pm$ 0.162	0.467 $\pm$ 0.153	0.525 $\pm$ 0.147	0.428 $\pm$ 0.145	0.399 $\pm$ 0.120	0.381 $\pm$ 0.118

Following prior studies (Bao et al., 2024), each dataset is randomly divided into training, validation, and testing subsets with a ratio of 8:1:1. All experimental evaluations reported below are conducted under this configuration for consistency.

4.1. Synthetic Data Evaluation

Data Generation. To rigorously evaluate the performance of our proposed ML-NAM, we conduct a simulation study following the experimental setup in Chen et al. (2020); Wang et al. (2025). The synthetic data are

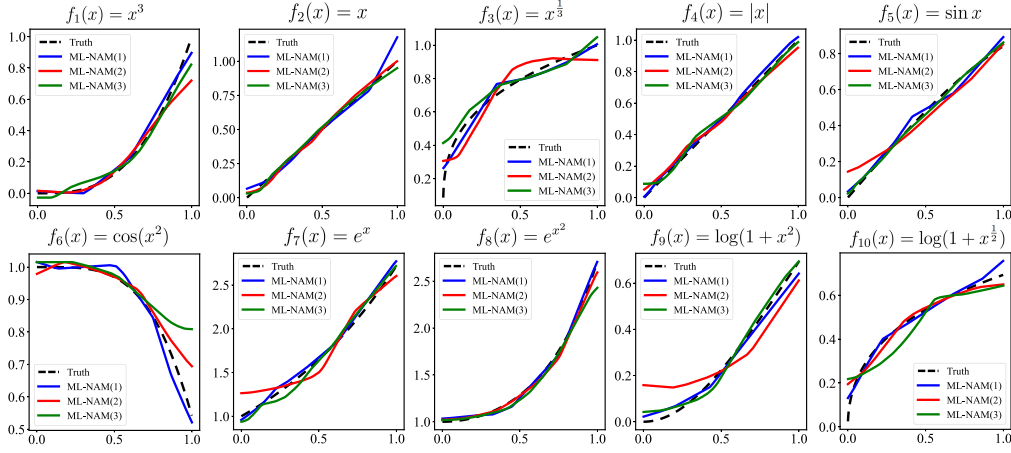


Figure 4: Estimated component functions obtained by ML-NAM under three noise settings without outliers. The figure shows the true functions (solid lines) and the corresponding ML-NAM estimates under (1) Standard Gaussian noise (ML-NAM(1)), (2) Mixture Gaussian noise (ML-NAM(2)), and (3) Student’s t-distribution noise (ML-NAM(3)).

generated from the model $y_i = f^*(\mathbf{x}_i) + \epsilon_i$, where the ground-truth function $f^*(\mathbf{x}_i)$ is an additive composition of ten distinct nonlinear functions:

$$f^*(\mathbf{x}_i) = \sum_{j=1}^{10} f_j^*(x_i^j) = (x_i^1)^3 + (x_i^2) + (x_i^3)^{\frac{1}{3}} + |x_i^4| + \sin(x_i^5) + \cos((x_i^6)^2) \\ + e^{(x_i^7)} + e^{(x_i^8)^2} + \log(1 + (x_i^9)^2) + \log(1 + (x_i^{10})^{\frac{1}{2}}).$$

The error term ϵ_i is drawn from three noise distributions: (1) Standard normal distribution: $\epsilon_i \sim \mathcal{N}(0, 1)$, (2) Mixture Gaussian distribution $\epsilon_i \sim 0.6\mathcal{N}(0, 1) + 0.4\mathcal{N}(0, 3)$, and (3) Student’s t-distribution $\epsilon_i \sim t(3)$. For each simulation, we generate a training set of $n = 1024$ samples, with covariates $\mathbf{x}_i$ drawn from a uniform distribution $\mathcal{U}([0, 1]^p)$. To assess the model’s scalability, we vary the input dimensionality $p \in \{10, 100, 200, 400\}$. In addition, we perturb the labels y of a fraction $r \in \{0\%, 10\%\}$ of the training samples by injecting an additional Gaussian noise term drawn from $\mathcal{N}(100, 100)$.

Visualization on Interpretability. To evaluate the robustness and accuracy of ML-NAM under various noise conditions, we visualize the estimated component functions alongside the actual functions in two distinct scenarios: one without outliers and the other with outliers. Figures 4 and 5 illustrate the estimated curves obtained under three types of noise distribu-

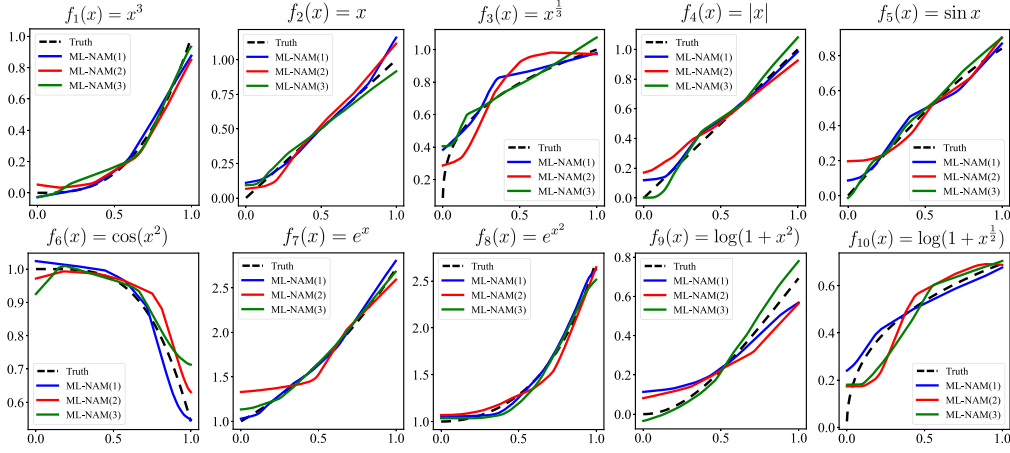


Figure 5: Estimated component functions obtained by ML-NAM under three noise settings with outliers. The figure shows the true functions (solid lines) and the corresponding ML-NAM estimates under (1) Standard Gaussian noise with outliers (ML-NAM(1)), (2) Mixture Gaussian noise with outliers (ML-NAM(2)), and (3) Student’s t-distribution with outliers (ML-NAM(3)).

tions. The first setting examines the performance of ML-NAM in the presence of clean noise, while the second setting incorporates additional outliers to test its robustness. These visualizations provide an intuitive comparison of the model’s estimation quality across different noise structures. This provides strong evidence for the model’s high interpretability and its ability to accurately capture the fine-grained structure of each feature’s contribution, even in the presence of significant noise and outliers.

Prediction Performance. Table 3 summarizes the prediction errors of ML-NAM against several baseline models, including MLP, NAM, NBM, and CAT. The results clearly indicate that while the performance of baseline methods degrades substantially under the non-Gaussian noise distributions, our ML-NAM demonstrates superior robustness. It consistently achieves lower prediction errors across all experimental settings, confirming its effectiveness and reliability in challenging data environments.

To quantitatively assess the improvement in shape estimation, we calculated the MAE for each shape function in Table 4. The results indicate that the MAE remains consistently low, even for rapidly varying functions, confirming that the estimated curves closely match the ground truth. Notably, in the presence of 10% outliers, models based on heavy-tailed distributions

Table 4: Experimental results of the shape functions (average MAE ($\downarrow$) $\pm$ standard deviation) on the simulation datasets. The left and right panels summarize the performance under outlier ratios of $r = 0\%$ and $r = 10\%$, respectively.

Shape Functions	Outlier Ratio $r = 0\%$			Outlier Ratio $r = 10\%$		
	Normal Gaussian	Gaussian Mixture	Student-t	Normal Gaussian	Gaussian Mixture	Student-t
$f_1(x) = x^3$	0.024 ± 0.006	0.034 ± 0.001	0.030 ± 0.007	0.021 ± 0.005	0.026 ± 0.009	0.029 ± 0.002
$f_2(x) = x$	0.025 ± 0.006	0.019 ± 0.003	0.018 ± 0.010	0.037 ± 0.002	0.029 ± 0.006	0.015 ± 0.006
$f_3(x) = x^{\frac{1}{3}}$	0.026 ± 0.002	0.048 ± 0.003	0.033 ± 0.007	0.030 ± 0.003	0.054 ± 0.015	0.037 ± 0.002
$f_4(x) = x $	0.013 ± 0.001	0.033 ± 0.004	0.034 ± 0.006	0.020 ± 0.004	0.029 ± 0.015	0.030 ± 0.004
$f_5(x) = \sin x$	0.023 ± 0.007	0.041 ± 0.011	0.016 ± 0.004	0.024 ± 0.004	0.045 ± 0.013	0.019 ± 0.004
$f_6(x) = \cos(x^2)$	0.028 ± 0.002	0.029 ± 0.013	0.036 ± 0.006	0.032 ± 0.001	0.028 ± 0.012	0.032 ± 0.005
$f_7(x) = e^x$	0.013 ± 0.003	0.055 ± 0.008	0.029 ± 0.012	0.021 ± 0.004	0.056 ± 0.005	0.022 ± 0.004
$f_8(x) = e^{x^2}$	0.034 ± 0.008	0.031 ± 0.008	0.028 ± 0.005	0.031 ± 0.002	0.035 ± 0.007	0.027 ± 0.004
$f_9(x) = \log(1 + x^2)$	0.028 ± 0.010	0.042 ± 0.022	0.035 ± 0.007	0.026 ± 0.006	0.048 ± 0.010	0.042 ± 0.008
$f_{10}(x) = \log(1 + x^{\frac{1}{2}})$	0.012 ± 0.002	0.019 ± 0.004	0.034 ± 0.001	0.015 ± 0.002	0.034 ± 0.004	0.036 ± 0.003

such as the Student-t demonstrate stronger robustness. This suggests that an appropriate likelihood specification enables the framework to model complex structures accurately while effectively mitigating the influence of outliers.

4.2. Real World Data Evaluation

Data Source and Description. Now, we validate the practical utility and robustness of ML-NAM through experiments on four publicly available benchmark datasets. The SkillCraft Master Table dataset (SCMT, $n = 3338$, $p = 19$) comprises gameplay statistics from StarCraft II players, where the task is to predict player skill levels based on behavioral and performance features (Thompson et al., 2013). The Gas Sensor Array Drift dataset (GSAD, $n = 2565$, $p = 128$) contains time-series recordings from an array of 16 chemical sensors collected over three years, to estimate gas component concentrations under sensor drift conditions (Vergara, 2012). The Coronal Mass Ejection (CME, downloaded from <https://cdaw.gsfc.nasa.gov/CMElist/>) dataset ($n = 137$, $p = 21$) provides event-level measurements of CME properties, aiming to predict their arrival times (Liu et al., 2018). The Alzheimer’s Disease Neuroimaging Initiative (ADNI) dataset ($n = 795$, $p = 326$, see adni.loni.usc.edu) includes multimodal clinical, imaging, and biomarker data, with the task of predicting the progression of Alzheimer’s disease. Furthermore, the extreme outliers are generated by corrupting a fraction $r \in \{0\%, 10\%, 30\%\}$ of training samples with additional Gaussian noise $\mathcal{N}(100, 100)$ in their labels y .

Prediction Performance. Table 5 presents the prediction results on real-world datasets. The findings show that our ML-NAM consistently achieves

Table 5: Experimental results on the real-world datasets, where the upper and lower panels report the average MSE ($\downarrow$) and average MAE ($\downarrow$) with their standard deviations, respectively. Here, r denotes the proportion of noisy labels (outliers). The best results among all baselines are highlighted in **bold**.

Datasets	r	MLP	XGBoost	NAM	NBM	CAT	ML-NAM	ML-NAM ⁺
SCMT	0%	0.030 $\pm$ 0.017	0.008 $\pm$ 0.005	0.024 $\pm$ 0.012	0.013 $\pm$ 0.009	0.029 $\pm$ 0.017	0.012 $\pm$ 0.007	0.010 $\pm$ 0.003
	10%	0.656 $\pm$ 0.054	0.651 $\pm$ 0.052	0.597 $\pm$ 0.044	0.632 $\pm$ 0.058	0.588 $\pm$ 0.031	0.447 $\pm$ 0.025	0.443 $\pm$ 0.017
	30%	1.145 $\pm$ 0.123	1.108 $\pm$ 0.111	1.086 $\pm$ 0.118	1.029 $\pm$ 0.113	0.984 $\pm$ 0.098	0.728 $\pm$ 0.085	0.679 $\pm$ 0.074
GSAD	0%	0.433 $\pm$ 0.133	0.423 $\pm$ 0.116	0.430 $\pm$ 0.129	0.418 $\pm$ 0.112	0.427 $\pm$ 0.121	0.411 $\pm$ 0.106	0.407 $\pm$ 0.098
	10%	1.183 $\pm$ 0.154	1.033 $\pm$ 0.143	0.922 $\pm$ 0.137	0.940 $\pm$ 0.136	0.904 $\pm$ 0.123	0.684 $\pm$ 0.121	0.646 $\pm$ 0.115
	30%	1.313 $\pm$ 0.212	1.281 $\pm$ 0.201	1.032 $\pm$ 0.207	1.289 $\pm$ 0.231	0.991 $\pm$ 0.196	0.795 $\pm$ 0.163	0.786 $\pm$ 0.154
CME	0%	0.598 $\pm$ 0.129	0.604 $\pm$ 0.140	0.619 $\pm$ 0.118	0.613 $\pm$ 0.098	0.585 $\pm$ 0.107	0.591 $\pm$ 0.116	0.579 $\pm$ 0.103
	10%	0.863 $\pm$ 0.217	0.844 $\pm$ 0.194	0.874 $\pm$ 0.199	0.867 $\pm$ 0.159	0.830 $\pm$ 0.155	0.627 $\pm$ 0.139	0.614 $\pm$ 0.117
	30%	1.611 $\pm$ 0.743	1.581 $\pm$ 0.726	1.603 $\pm$ 0.481	1.584 $\pm$ 0.459	1.310 $\pm$ 0.422	0.721 $\pm$ 0.192	0.694 $\pm$ 0.162
ADNI	0%	0.174 $\pm$ 0.042	0.167 $\pm$ 0.037	0.190 $\pm$ 0.048	0.182 $\pm$ 0.041	0.175 $\pm$ 0.045	0.184 $\pm$ 0.059	0.172 $\pm$ 0.041
	10%	0.342 $\pm$ 0.091	0.336 $\pm$ 0.094	0.372 $\pm$ 0.114	0.364 $\pm$ 0.108	0.324 $\pm$ 0.092	0.251 $\pm$ 0.073	0.218 $\pm$ 0.074
	30%	0.746 $\pm$ 0.139	0.739 $\pm$ 0.128	0.713 $\pm$ 0.132	0.706 $\pm$ 0.119	0.649 $\pm$ 0.102	0.313 $\pm$ 0.114	0.293 $\pm$ 0.091
Datasets	r	MLP	XGBoost	NAM	NBM	CAT	ML-NAM	ML-NAM ⁺
SCMT	0%	0.131 $\pm$ 0.009	0.032 $\pm$ 0.004	0.087 $\pm$ 0.006	0.069 $\pm$ 0.003	0.096 $\pm$ 0.008	0.066 $\pm$ 0.004	0.048 $\pm$ 0.002
	10%	0.429 $\pm$ 0.039	0.421 $\pm$ 0.037	0.403 $\pm$ 0.033	0.436 $\pm$ 0.025	0.401 $\pm$ 0.018	0.375 $\pm$ 0.014	0.343 $\pm$ 0.011
	30%	0.644 $\pm$ 0.062	0.632 $\pm$ 0.054	0.626 $\pm$ 0.058	0.586 $\pm$ 0.052	0.573 $\pm$ 0.043	0.486 $\pm$ 0.035	0.454 $\pm$ 0.032
GSAD	0%	0.380 $\pm$ 0.061	0.362 $\pm$ 0.053	0.334 $\pm$ 0.046	0.346 $\pm$ 0.046	0.321 $\pm$ 0.042	0.294 $\pm$ 0.041	0.286 $\pm$ 0.039
	10%	0.502 $\pm$ 0.087	0.491 $\pm$ 0.079	0.413 $\pm$ 0.081	0.422 $\pm$ 0.089	0.407 $\pm$ 0.076	0.323 $\pm$ 0.056	0.318 $\pm$ 0.045
	30%	0.718 $\pm$ 0.162	0.703 $\pm$ 0.147	0.698 $\pm$ 0.141	0.724 $\pm$ 0.156	0.687 $\pm$ 0.143	0.617 $\pm$ 0.138	0.605 $\pm$ 0.131
CME	0%	0.474 $\pm$ 0.113	0.477 $\pm$ 0.121	0.487 $\pm$ 0.108	0.485 $\pm$ 0.099	0.465 $\pm$ 0.104	0.469 $\pm$ 0.106	0.462 $\pm$ 0.102
	10%	0.730 $\pm$ 0.147	0.718 $\pm$ 0.140	0.737 $\pm$ 0.141	0.731 $\pm$ 0.126	0.711 $\pm$ 0.125	0.591 $\pm$ 0.118	0.583 $\pm$ 0.108
	30%	1.072 $\pm$ 0.662	1.060 $\pm$ 0.652	1.066 $\pm$ 0.395	1.057 $\pm$ 0.376	0.945 $\pm$ 0.351	0.649 $\pm$ 0.238	0.633 $\pm$ 0.202
ADNI	0%	0.118 $\pm$ 0.063	0.109 $\pm$ 0.058	0.136 $\pm$ 0.068	0.127 $\pm$ 0.066	0.118 $\pm$ 0.067	0.130 $\pm$ 0.070	0.115 $\pm$ 0.062
	10%	0.285 $\pm$ 0.096	0.280 $\pm$ 0.094	0.310 $\pm$ 0.107	0.303 $\pm$ 0.108	0.270 $\pm$ 0.096	0.262 $\pm$ 0.073	0.257 $\pm$ 0.077
	30%	0.664 $\pm$ 0.118	0.659 $\pm$ 0.112	0.644 $\pm$ 0.115	0.643 $\pm$ 0.109	0.605 $\pm$ 0.105	0.402 $\pm$ 0.106	0.354 $\pm$ 0.099

lower MSE and standard deviation than the competing methods, particularly as the proportion of outliers increases ($r = 10\%$ and 30%). It is noteworthy that in the absence of outliers ($r = 0\%$), XGBoost occasionally exhibits a marginal advantage, attributable to its powerful approximation capabilities derived from ensembling multiple weak learners. However, its performance degrades more significantly than ML-NAM's in the presence of data contamination. Overall, these empirical evaluations on real-world data corroborate the findings from our simulation study, consistently demonstrating the superior robustness and accuracy of ML-NAM in environments with non-Gaussian noise and extreme outliers.

5. Ablations and Extensions

5.1. Available Kernels for KDE

To better demonstrate the flexibility of ML-NAM, we conduct extended experiments on the ADNI dataset using alternative KDE kernels, including the Gaussian, Exponential, Epanechnikov, Uniform, Sigmoid, Triangular, and Logistic kernels. The definitions are formulated as follows:

Table 6: Experimental results on the SCMT and ADNI datasets, with the upper panel reporting the average MSE ($\downarrow$) $\pm$ standard deviation and the lower panel presenting the corresponding average MAE ($\downarrow$), for evaluating different kernels in the KDE of ML-NAM.

Datasets	r	Kernel Type						
		Gaussian	Exponential	Epanechnikov	Uniform	Sigmoid	Triangular	Logistic
SCMT	0%	0.012 $\pm$ 0.007	0.016 $\pm$ 0.011	0.013 $\pm$ 0.009	0.018 $\pm$ 0.013	0.016 $\pm$ 0.012	0.014 $\pm$ 0.008	0.017 $\pm$ 0.014
	10%	0.447 $\pm$ 0.025	0.433 $\pm$ 0.021	0.452 $\pm$ 0.033	0.458 $\pm$ 0.037	0.450 $\pm$ 0.029	0.436 $\pm$ 0.023	0.438 $\pm$ 0.026
	30%	0.728 $\pm$ 0.085	0.732 $\pm$ 0.088	0.745 $\pm$ 0.091	0.753 $\pm$ 0.122	0.748 $\pm$ 0.113	0.746 $\pm$ 0.097	0.741 $\pm$ 0.092
ADNI	0%	0.184 $\pm$ 0.059	0.192 $\pm$ 0.062	0.201 $\pm$ 0.065	0.215 $\pm$ 0.071	0.207 $\pm$ 0.068	0.198 $\pm$ 0.064	0.195 $\pm$ 0.063
	10%	0.251 $\pm$ 0.073	0.267 $\pm$ 0.078	0.281 $\pm$ 0.082	0.298 $\pm$ 0.089	0.286 $\pm$ 0.084	0.275 $\pm$ 0.080	0.271 $\pm$ 0.079
	30%	0.313 $\pm$ 0.114	0.338 $\pm$ 0.122	0.356 $\pm$ 0.131	0.382 $\pm$ 0.143	0.364 $\pm$ 0.135	0.347 $\pm$ 0.127	0.342 $\pm$ 0.125

Datasets	r	Kernel Type						
		Gaussian	Exponential	Epanechnikov	Uniform	Sigmoid	Triangular	Logistic
SCMT	0%	0.066 $\pm$ 0.004	0.074 $\pm$ 0.008	0.068 $\pm$ 0.007	0.073 $\pm$ 0.008	0.070 $\pm$ 0.009	0.068 $\pm$ 0.005	0.073 $\pm$ 0.010
	10%	0.375 $\pm$ 0.014	0.362 $\pm$ 0.012	0.385 $\pm$ 0.018	0.389 $\pm$ 0.020	0.382 $\pm$ 0.017	0.367 $\pm$ 0.014	0.370 $\pm$ 0.015
	30%	0.486 $\pm$ 0.035	0.492 $\pm$ 0.037	0.499 $\pm$ 0.038	0.506 $\pm$ 0.051	0.503 $\pm$ 0.047	0.498 $\pm$ 0.043	0.496 $\pm$ 0.039
ADNI	0%	0.130 $\pm$ 0.070	0.138 $\pm$ 0.076	0.149 $\pm$ 0.077	0.164 $\pm$ 0.076	0.155 $\pm$ 0.072	0.145 $\pm$ 0.072	0.142 $\pm$ 0.070
	10%	0.262 $\pm$ 0.073	0.271 $\pm$ 0.079	0.292 $\pm$ 0.075	0.304 $\pm$ 0.078	0.298 $\pm$ 0.073	0.284 $\pm$ 0.079	0.282 $\pm$ 0.079
	30%	0.402 $\pm$ 0.106	0.482 $\pm$ 0.110	0.496 $\pm$ 0.115	0.519 $\pm$ 0.120	0.504 $\pm$ 0.118	0.490 $\pm$ 0.113	0.485 $\pm$ 0.111

- 501 • Gaussian kernel: $K(u, u') = \exp\left(-\frac{\|u-u'\|^2}{2\sigma^2}\right)$,
- 502 • Exponential kernel: $K(u, u') = \exp\left(-\frac{\|u-u'\|}{\sigma}\right)$,
- 503 • Epanechnikov kernel: $K(u, u') = \max\left(0, 1 - \frac{\|u-u'\|^2}{\sigma^2}\right)$,
- 504 • Uniform kernel: $K(u, u') = \mathbb{I}(\|u - u'\| \leq \sigma)$,
- 505 • Sigmoid kernel: $K(u, u') = \frac{1}{1 + \exp\left(-\frac{\|u-u'\|}{\sigma}\right)}$,
- 506 • Triangular kernel: $K(u, u') = \max\left(0, 1 - \frac{\|u-u'\|}{\sigma}\right)$,
- 507 • Logistic kernel: $K(u, u') = \frac{2}{\exp\left(\frac{\|u-u'\|}{\sigma}\right) + \exp\left(-\frac{\|u-u'\|}{\sigma}\right)}$,

508 where u and u' denote the two inputs of kernels for distance-based similarity
509 evaluation.

510 **Experimental Setup:** We evaluate ML-NAM with different kernels on
511 the SCMT and ADNI datasets under three outlier scenarios ($r \in \{0\%, 10\%, 30\%\}$).
512 For each kernel, we optimize the bandwidth parameter σ using 5-fold cross-
513 validation from the range $\{0.01, 0.05, 0.1, 0.5\}$. All other hyperparameters
514 remain consistent with the previous experiments. Each configuration is re-
515 peated 30 times to ensure statistical significance.

Results: The performance of ML-NAM with different kernels is presented in Table 6. Table 6 demonstrates the flexibility of ML-NAM in accommodating different kernel functions for density estimation. The Gaussian kernel achieves the best performance across all outlier scenarios, likely due to its smoothness properties and ability to capture complex density distributions. However, even with less common kernels, such as the Uniform and Sigmoid kernels, ML-NAM maintains reasonable performance, highlighting the robustness of our maximum likelihood framework. This flexibility is particularly valuable in real-world applications where the underlying noise distribution may not be well-characterized, allowing practitioners to select the most appropriate kernel based on domain knowledge or computational considerations.

5.2. Bandwidth Sensitivity Analysis

To assess the influence of kernel bandwidth on model performance, we conduct an ablation study on the SCMT dataset, considering both Gaussian and exponential kernel functions.

Experimental Setup: We vary σ within the range $\{0.01, 0.05, 0.1, 0.5\}$. All remaining hyperparameters follow the same configuration used in the baseline experiments to isolate the impact of kernel bandwidth. Each configuration is independently run 30 times to ensure statistical reliability and reduce variance in performance estimates.

Table 7: Experimental results reporting the average MSE ($\downarrow$) $\pm$ standard deviation in the left panel and the corresponding average MAE ($\downarrow$) $\pm$ standard deviation in the right panel for different kernels and bandwidth values σ .

Kernel	r	MSE				MAE			
		$\sigma = 0.01$	$\sigma = 0.05$	$\sigma = 0.1$	$\sigma = 0.5$	$\sigma = 0.01$	$\sigma = 0.05$	$\sigma = 0.1$	$\sigma = 0.5$
Gaussian	0%	0.015 $\pm$ 0.008	0.012 $\pm$ 0.007	0.026 $\pm$ 0.017	0.089 $\pm$ 0.023	0.069 $\pm$ 0.006	0.066 $\pm$ 0.004	0.102 $\pm$ 0.016	0.113 $\pm$ 0.034
	10%	0.447 $\pm$ 0.025	0.451 $\pm$ 0.028	0.470 $\pm$ 0.054	0.625 $\pm$ 0.072	0.375 $\pm$ 0.014	0.382 $\pm$ 0.016	0.467 $\pm$ 0.053	0.516 $\pm$ 0.078
	30%	0.731 $\pm$ 0.088	0.728 $\pm$ 0.085	0.798 $\pm$ 0.159	0.854 $\pm$ 0.204	0.494 $\pm$ 0.041	0.486 $\pm$ 0.035	0.628 $\pm$ 0.141	0.713 $\pm$ 0.187
Exponential	0%	0.016 $\pm$ 0.011	0.018 $\pm$ 0.010	0.028 $\pm$ 0.019	0.092 $\pm$ 0.026	0.074 $\pm$ 0.008	0.077 $\pm$ 0.009	0.087 $\pm$ 0.017	0.103 $\pm$ 0.038
	10%	0.441 $\pm$ 0.023	0.433 $\pm$ 0.021	0.474 $\pm$ 0.052	0.629 $\pm$ 0.075	0.371 $\pm$ 0.015	0.362 $\pm$ 0.012	0.489 $\pm$ 0.054	0.512 $\pm$ 0.076
	30%	0.737 $\pm$ 0.092	0.732 $\pm$ 0.088	0.796 $\pm$ 0.158	0.863 $\pm$ 0.208	0.496 $\pm$ 0.043	0.492 $\pm$ 0.037	0.632 $\pm$ 0.144	0.719 $\pm$ 0.191

Results: The results in Table 7 show that the bandwidth σ substantially affects the performance of ML-NAM. Smaller bandwidths enable the model to capture local variations more effectively, yielding lower prediction errors, whereas larger values introduce noticeable over-smoothing and degrade accuracy. Moreover, ML-NAM remains robust to noise: even under a 30% noise level, the configuration with $\sigma = 0.01$ maintains stable performance, demonstrating its ability to mitigate outliers. Overall, a small bandwidth provides

the most favorable balance between accuracy and robustness, consistent with the theoretical insights in Corollary 2.

5.3. Computational Efficiency Analysis

To assess the computational efficiency of ML-NAM, we conduct experiments comparing training time and memory consumption against baseline models across different dataset sizes and dimensionalities, utilizing the synthetic dataset in Section 4.1.

Experimental Setup: We evaluate ML-NAM against NAM, NBM, and CAT on synthetic datasets with varying datasizes ($n \in \{512, 1024, 2048, 4096\}$) and feature dimensions ($p \in \{10, 50, 100, 200\}$). The dimension is modified by adding noisy features following the distribution of $\mathcal{N}(0, 1)$. All experiments are conducted on a workstation with the same hardware and software settings. We measure the training time (in seconds) for each model, using the same hyperparameter settings as those in Section 4. Each configuration is repeated 10 times to ensure consistency and reliability.

Table 8: Computational efficiency comparison on training time (seconds) across different dataset sizes and dimensions.

(n, p)	Model			
	NAM	NBM	CAT	ML-NAM
(512, 10)	12.3 ± 1.2	15.7 ± 1.5	10.8 ± 1.0	14.2 ± 1.3
(1024, 50)	28.6 ± 2.1	35.2 ± 2.8	24.7 ± 1.9	32.4 ± 2.5
(2048, 100)	67.3 ± 4.2	82.5 ± 5.1	58.9 ± 3.8	75.8 ± 4.7
(4096, 200)	152.4 ± 8.3	187.6 ± 9.7	134.2 ± 7.5	171.3 ± 8.9

Results: The computational efficiency results are presented in Table 8. The results in Table 8 demonstrate that ML-NAM maintains competitive computational efficiency despite its additional density estimation step. The training time of ML-NAM is approximately 15 – 20% higher than that of standard NAM, which is a reasonable trade-off given the significant improvements in robustness and accuracy demonstrated in previous experiments.

This analysis confirms that ML-NAM can be applied to large-scale problems without prohibitive computational overhead, especially for real-world applications where both robustness and efficiency are critical. These practical comparisons highlight ML-NAM’s capabilities in robust regression tasks, including handling outliers and non-Gaussian noise that commonly degrade model performance in real-world settings.

5.4. Extension to Group-wise Modelling

To further enhance the interpretability and practical utility of ML-NAM, we extend our framework to support group-wise modeling, where input features can be organized into predefined groups based on domain knowledge.

Modified Structure and Objects. We slightly modify the initial model structure in Figure 2 to accommodate the group-wise feature selection task better. As presented in Figure 6, the modified ML-NAM with group lasso penalty (Bach, 2008) is denoted as ML-NAM^g. Furthermore, inspired by Sparse Group Lasso (Simon et al., 2013), we also construct the sparse ML-NAM with both group-wise $\ell_{2,1}$ and individual feature-wise ℓ_2 penalties, denoted by ML-NAM^{g+}. This extension is particularly valuable in applications like genomics (gene pathways) or healthcare (related clinical measurements), where features within the same group often share similar characteristics or biological functions.

Experimental Setup: We evaluate the group-wise extension of ML-NAM (denoted as ML-NAM^g) on the SCMT and ADNI clinical datasets. To formulate the group modeling framework, we define feature groups based on the structural categories of the variables or prior domain knowledge. For the SCMT dataset, we followed the approach in Thompson et al. (2013) and divided these features into three categories: Player Profile, Physical Ability, and Strategy. And for the ADNI dataset, features are partitioned according to its prior known group structure (Zhang et al., 2023). The dataset comprises 116 variable groups: 46 groups contain a single covariate (subcortical volume, SV), while the remaining 70 groups each include four covariates: cortical thickness average (TA), thickness standard deviation (TS), surface area (SA), and cortical volume (CV). The inherent grouping structure provides a valuable framework for our analytical approach and serves as a fundamental organizational principle for the neuroimaging data utilized in this investigation.

We compare ML-NAM^g with standard ML-NAM and other group modeling approaches, such as Group Lasso and Sparse Group Lasso. For ML-NAM^g, we implement a group sparsity penalty that encourages feature selection at the group level while maintaining the maximum likelihood estimation framework. The regularization strength is tuned via 5-fold cross-validation from $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$. Experiments are conducted under three outlier scenarios ($r \in \{0\%, 10\%, 30\%\}$), with each configuration repeated 30 times.

Results. The performance of different group modeling approaches is presented in Table 9. Our ML-NAM^{g+} (with ℓ_2 penalty for single feature se-

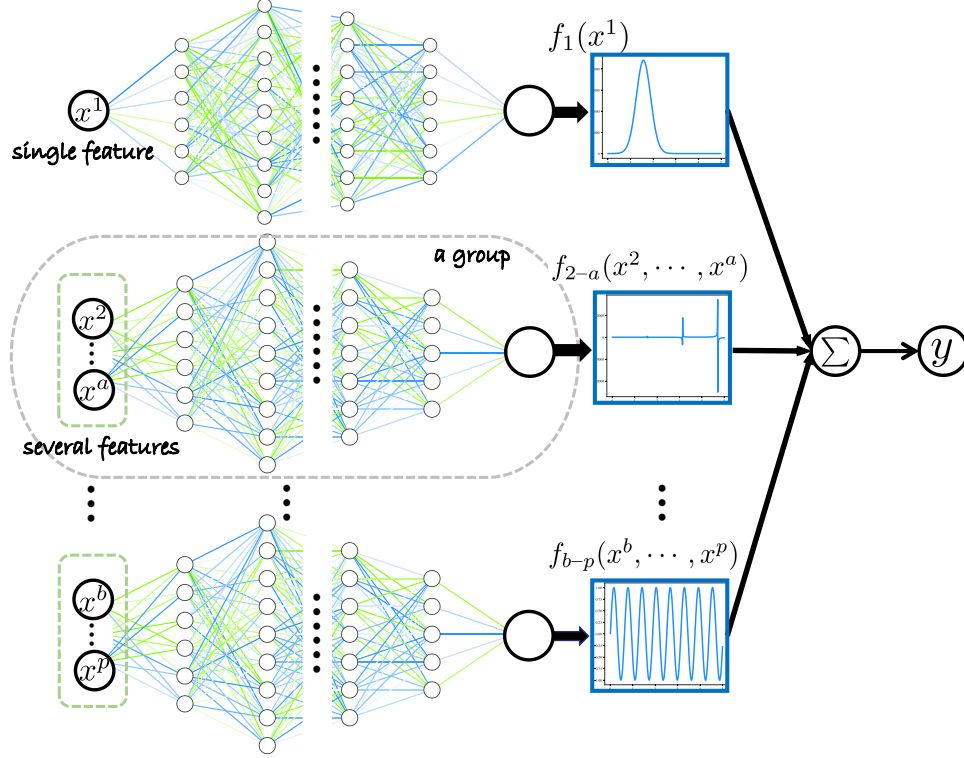


Figure 6: Structure of group version of ML-NAM (ML-NAM^g and ML-NAM^{g+}). ML-NAM^g is designed with $\ell_{2,1}$ sparse penalty for screening each group. ML-NAM^{g+} is the advanced ML-NAM^g equipped with the ℓ_2 penalty for individual feature selection. Grouped features are fed into the same subnetwork.

609 lection and $\ell_{2,1}$ penalty for group-wise penalty) achieves the best performance
 610 across all outlier scenarios, demonstrating the effectiveness of incorporating
 611 group structure into the maximum likelihood framework.

612 5.5. Summary on Experimental Findings

613 The experimental evaluation demonstrates that ML-NAM consistently
 614 outperforms existing methods across both synthetic and real-world datasets,
 615 particularly in the presence of non-Gaussian noise and outliers. On synthetic
 616 data, ML-NAM reduced MSE by 14%-29% compared to standard NAM under
 617 various non-Gaussian noise conditions, including heavy-tailed distributions
 618 and contaminated samples. This improvement was more pronounced
 619 as the dimensionality increased, highlighting the model's scalability and ro-

Table 9: Experimental results on the SCMT and ADNI datasets, with the upper panel reporting the average MSE ($\downarrow$) $\pm$ standard deviation and the lower panel presenting the corresponding average MAE ($\downarrow$), for validating group-wise modeling capabilities.

Datasets	r	Models					
		Group Lasso	Sparse Group Lasso	ML-NAM	ML-NAM ⁺	ML-NAM ^g	ML-NAM ^{g+}
SCMT	0%	0.011 $\pm$ 0.004	0.015 $\pm$ 0.003	0.012 $\pm$ 0.007	0.010 $\pm$ 0.003	0.009 $\pm$ 0.005	0.007 $\pm$ 0.003
	10%	0.514 $\pm$ 0.036	0.506 $\pm$ 0.029	0.447 $\pm$ 0.025	0.443 $\pm$ 0.017	0.441 $\pm$ 0.019	0.436 $\pm$ 0.014
	30%	0.982 $\pm$ 0.101	0.956 $\pm$ 0.092	0.728 $\pm$ 0.085	0.679 $\pm$ 0.074	0.674 $\pm$ 0.071	0.652 $\pm$ 0.066
ADNI	0%	0.180 $\pm$ 0.047	0.190 $\pm$ 0.048	0.184 $\pm$ 0.059	0.172 $\pm$ 0.041	0.176 $\pm$ 0.043	0.168 $\pm$ 0.039
	10%	0.385 $\pm$ 0.116	0.372 $\pm$ 0.114	0.251 $\pm$ 0.073	0.218 $\pm$ 0.074	0.237 $\pm$ 0.069	0.205 $\pm$ 0.065
	30%	0.731 $\pm$ 0.136	0.713 $\pm$ 0.132	0.313 $\pm$ 0.114	0.293 $\pm$ 0.091	0.298 $\pm$ 0.105	0.276 $\pm$ 0.087

Datasets	r	Models					
		Group Lasso	Sparse Group Lasso	ML-NAM	ML-NAM ⁺	ML-NAM ^g	ML-NAM ^{g+}
SCMT	0%	0.064 $\pm$ 0.003	0.071 $\pm$ 0.001	0.066 $\pm$ 0.004	0.048 $\pm$ 0.002	0.046 $\pm$ 0.003	0.045 $\pm$ 0.002
	10%	0.482 $\pm$ 0.031	0.476 $\pm$ 0.027	0.375 $\pm$ 0.014	0.343 $\pm$ 0.011	0.341 $\pm$ 0.017	0.336 $\pm$ 0.012
	30%	0.886 $\pm$ 0.052	0.864 $\pm$ 0.048	0.486 $\pm$ 0.035	0.454 $\pm$ 0.032	0.449 $\pm$ 0.036	0.427 $\pm$ 0.021
ADNI	0%	0.134 $\pm$ 0.068	0.146 $\pm$ 0.069	0.130 $\pm$ 0.070	0.115 $\pm$ 0.062	0.122 $\pm$ 0.075	0.110 $\pm$ 0.073
	10%	0.321 $\pm$ 0.108	0.310 $\pm$ 0.107	0.262 $\pm$ 0.073	0.257 $\pm$ 0.077	0.260 $\pm$ 0.074	0.253 $\pm$ 0.071
	30%	0.655 $\pm$ 0.149	0.644 $\pm$ 0.144	0.402 $\pm$ 0.106	0.354 $\pm$ 0.099	0.364 $\pm$ 0.104	0.326 $\pm$ 0.096

bustness. The visualization of estimated component functions revealed that ML-NAM accurately captures the underlying shape functions, even under challenging noise conditions.

In real-world applications, ML-NAM exhibited superior performance across four benchmark datasets spanning diverse domains. Notably, when outlier proportions increased, ML-NAM maintained significantly lower prediction errors than baseline methods. Our ML-NAM can adapt to complex noise distributions and feature group structures without requiring explicit distributional assumptions proved particularly valuable in medical diagnosis and sensitive data analysis, where real-world noise often deviates from idealized Gaussian assumptions. These results demonstrate that ML-NAM is a promising approach for high-stakes applications that require reliable decision-making under uncertain conditions.

6. Conclusion

This paper introduces the Maximum Likelihood Neural Additive Model (ML-NAM), a novel framework that achieves robustness and interpretability simultaneously through maximum likelihood estimation. Our theoretical analysis establishes non-asymptotic upper bounds on excess risk under mild conditions, demonstrating that ML-NAM achieves a minimax convergence rate exhibiting polynomial decay when the target regression function lies in a Besov space, without requiring explicit assumptions about noise distribution.

641 Experimental evaluations on both synthetic and real-world datasets validate
642 ML-NAM’s superior empirical performance for robust and interpretable pre-
643 dictions under complex non-Gaussian noise.

644 In the future, it is interesting to further investigate the strategies for re-
645 ducing the computational complexity of KDE (e.g., via random Fourier fea-
646 tures and fast Gauss transform (Yang et al., 2003)) and extending ML-NAM
647 to time-series tasks. To further select an appropriate bandwidth for KDE,
648 some advanced approaches, such as Silverman’s Rule of Thumb (Silverman,
649 2018), will also be considered in future research. Moreover, it is meaningful
650 to extend its theoretical foundations to accommodate weaker assumptions,
651 while exploring applications in critical domains where both robustness and
652 interpretability are paramount.

653 7. Supplementary information

654 The complete implementation code for this research has been made pub-
655 licly available at Github.

656 All data used are publicly available. The SCMT dataset is available
657 at <https://zenodo.org/records/17829625>. The GSAD dataset is avail-
658 able at [https://archive.ics.uci.edu/dataset/224/gas+sensor+array+](https://archive.ics.uci.edu/dataset/224/gas+sensor+array+drift+dataset)
659 [drift+dataset](https://archive.ics.uci.edu/dataset/224/gas+sensor+array+drift+dataset). The CME dataset can be found at [https://cdaw.gsfc.](https://cdaw.gsfc.nasa.gov/CMElist/)
660 [nasa.gov/CMElist/](https://cdaw.gsfc.nasa.gov/CMElist/). The real ADNI dataset can be downloaded from [http:](http://adni.loni.usc.edu)
661 [//adni.loni.usc.edu](http://adni.loni.usc.edu).

662 8. Acknowledgements

663 This work is supported by the National Natural Science Foundation of
664 China (Grant Nos. 62376104 and 12426512).

665 9. Statements and Declarations

- 666 • **LLM Usage.** LLM tools were used during the editing (e.g., grammar,
667 spelling, word choice). And the authors are fully accountable for the
668 content.
- 669 • **Conflict of interest/Competing interests.** The authors declare no
670 Conflicts of interest.

671 All authors drafted the work or revised it critically for its content and
672 agreed to be accountable for all aspects of the work.

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